

# A control scheme to connect synthetic nervous system with spiking motoneurons to a biomimetic quadruped.

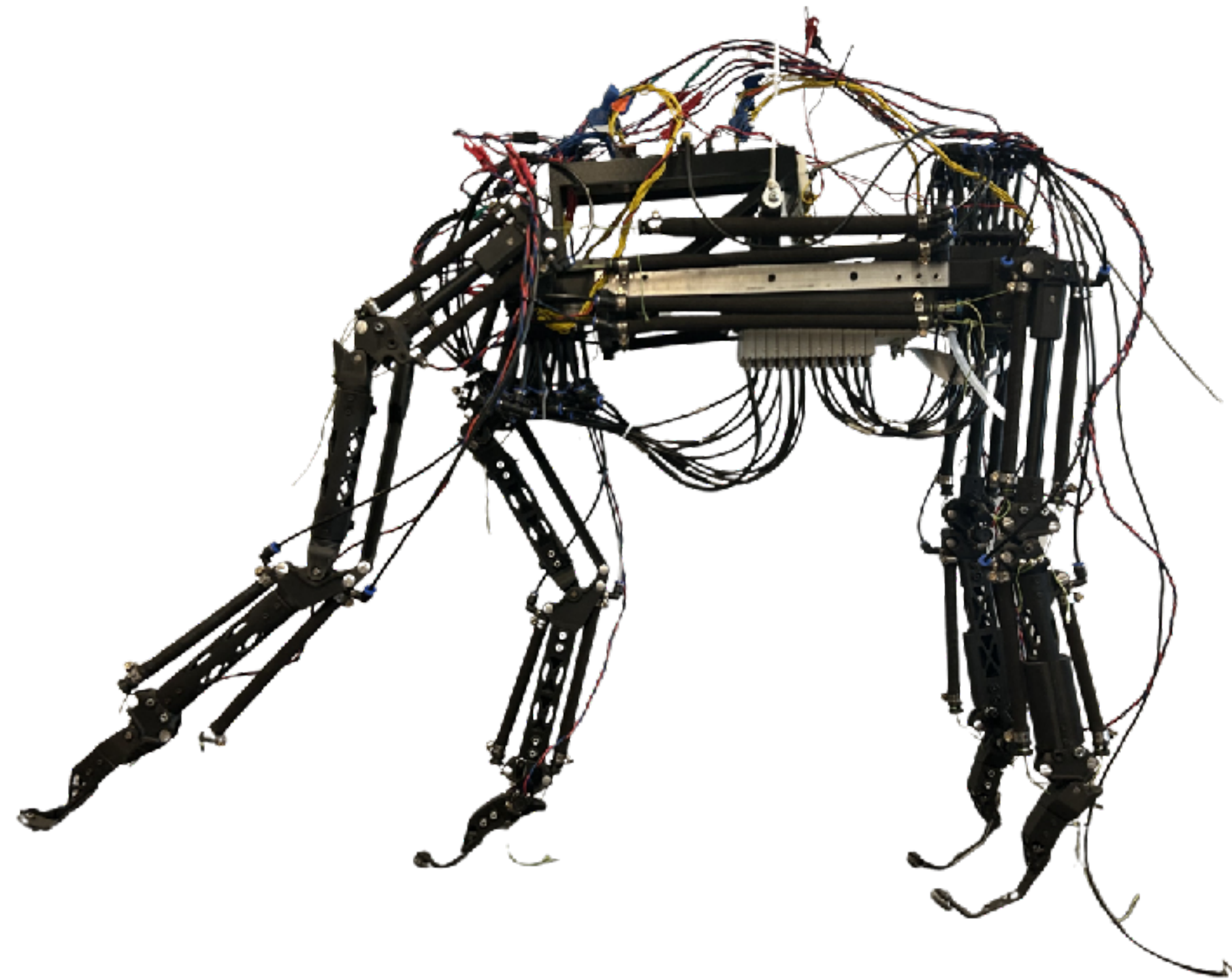


Photo by Simone Scheuer

## Thesis Defense

Matthew Jack Lutz

Dr. Alex Hunt



# Animals are extremely agile



<https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.akc.org%2Fexpert-advice%2Fnews%2Fdock-diving-whippet-guinness-wqwdmHWwXmL7hY&ust=1761850751053000&source=images&cd=vfe&opi=89978449&ved=0CByQjRxqFwoTjCJTsnLaYpADFQAAA>



[/dyeranimalclinic.com/blog/cat-jump/&psig=AOvWaw124Jy-jo&source=images&cd=vfe&opi=89978449&ved=0CBYQjRxqFwoTCLDztb-MypADFQAAAAAdAAAAABAE](http://dyeranimalclinic.com/blog/cat-jump/&psig=AOvWaw124Jy-jo&source=images&cd=vfe&opi=89978449&ved=0CBYQjRxqFwoTCLDztb-MypADFQAAAAAdAAAAABAE)



<https://www.google.com/url?sa=i&url=https://karenhoglundphotography.com/blog/portraits-of-dogs-jumping-in-water/&psig=AOvVaoi5FoaTf-qwdmHwWXML7hY&ust=1761850751053000&source=images&cd=vfe&opi=89978440&ved=0CBYQjRxqFwoTCJtJsNalypADfQAAAAAdAAAAABAI>



# Robots are... not as agile, yet!



[https://www.google.com/url?sa=i&url=https://www.mercurynews.com/2015/06/07/watch-a-compilation-of-robots-falling-down&psig=AOvVaw3LUmZaCCo5uwxrkglBulQO&ust=1761851049260000&source=images&cd=vfe&opi=89978449&ved=0CBYQjRxxqFwoTCCKDj\\_oGNypADFQAAAAAdAAAAABAK](https://www.google.com/url?sa=i&url=https://www.mercurynews.com/2015/06/07/watch-a-compilation-of-robots-falling-down&psig=AOvVaw3LUmZaCCo5uwxrkglBulQO&ust=1761851049260000&source=images&cd=vfe&opi=89978449&ved=0CBYQjRxxqFwoTCCKDj_oGNypADFQAAAAAdAAAAABAK)



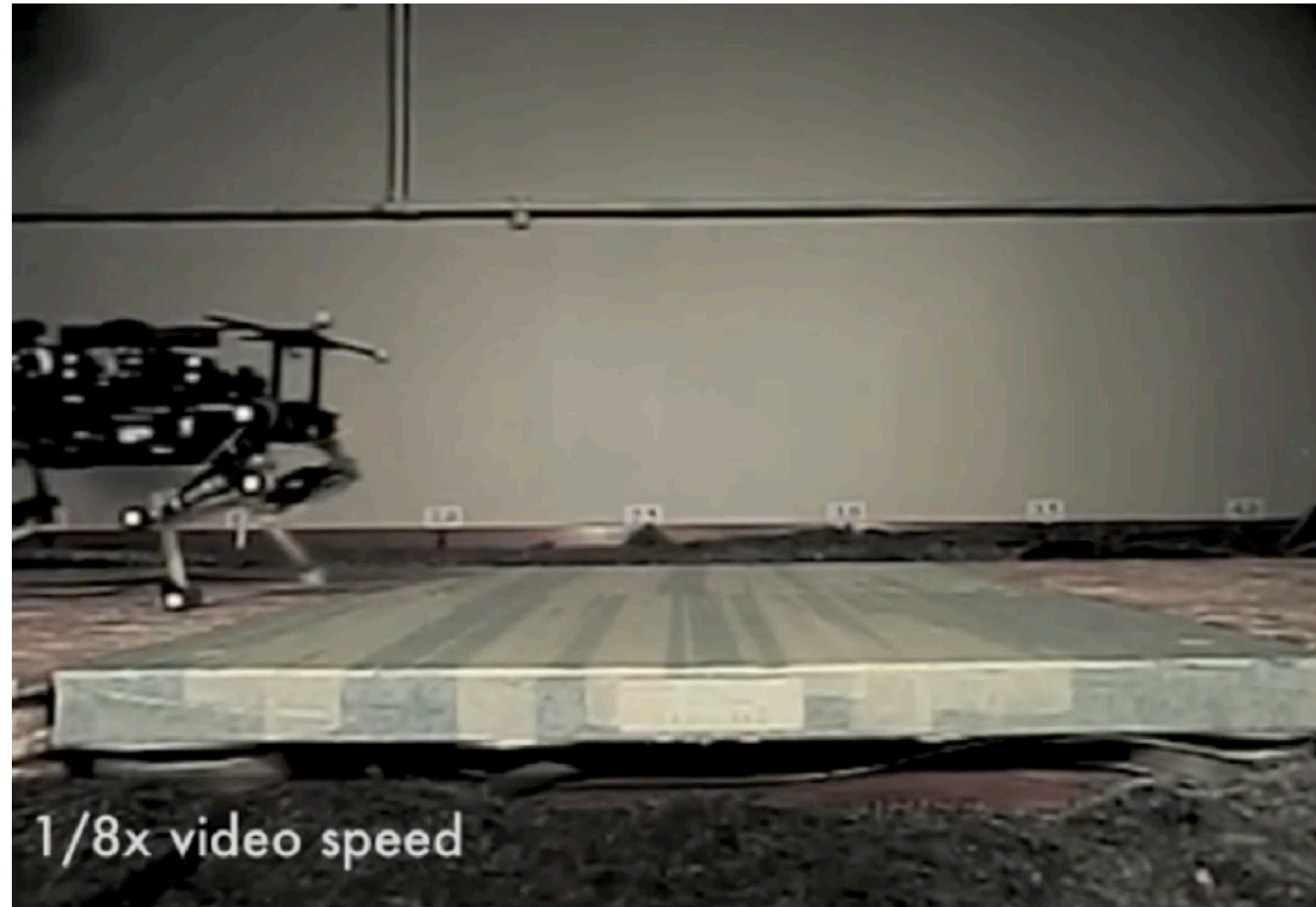
[https://www.google.com/url?sa=i&url=https://www.asu-team-unleashes-ai-powered-robot-dog-help-gestures&cd=vfe&opi=89978449&ved=0CBYQjRxxqFwoTCCKDj\\_oGNypADFQAAAAAdAAAAABAK](https://www.google.com/url?sa=i&url=https://www.asu-team-unleashes-ai-powered-robot-dog-help-gestures&cd=vfe&opi=89978449&ved=0CBYQjRxxqFwoTCCKDj_oGNypADFQAAAAAdAAAAABAK)



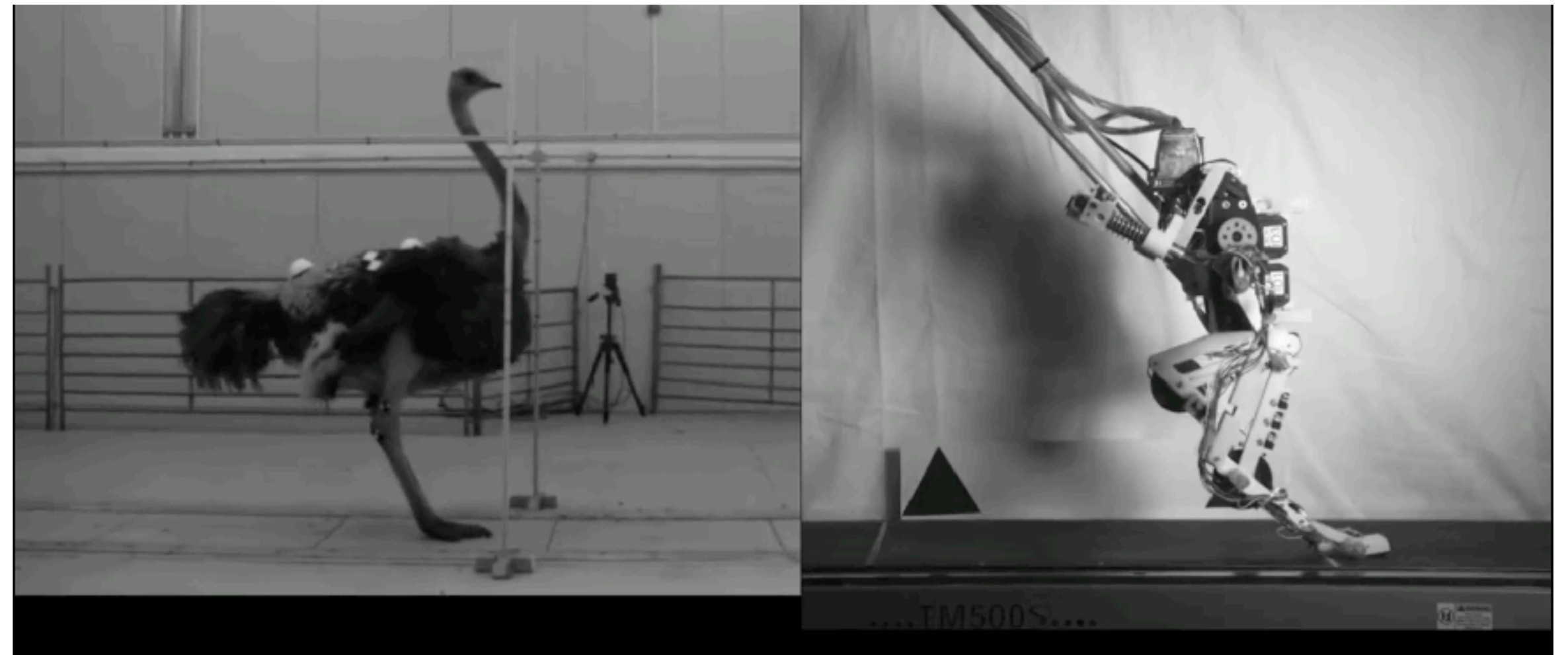
<https://www.google.com/url?sa=i&url=https://spectrum.ieee.org/darpa-robotics-challenge-robots-falling&psig=AOvVaw3Dange2puUGVcEVE3gtevs&ust=1761851205730000&source=images&cd=vfe&opi=89978449&ved=0CBYQjRxxqFwoTCNik7ayNypADFQAAAAAdAAAAABAE>



# Biomimetic robots attempt to learn from biology



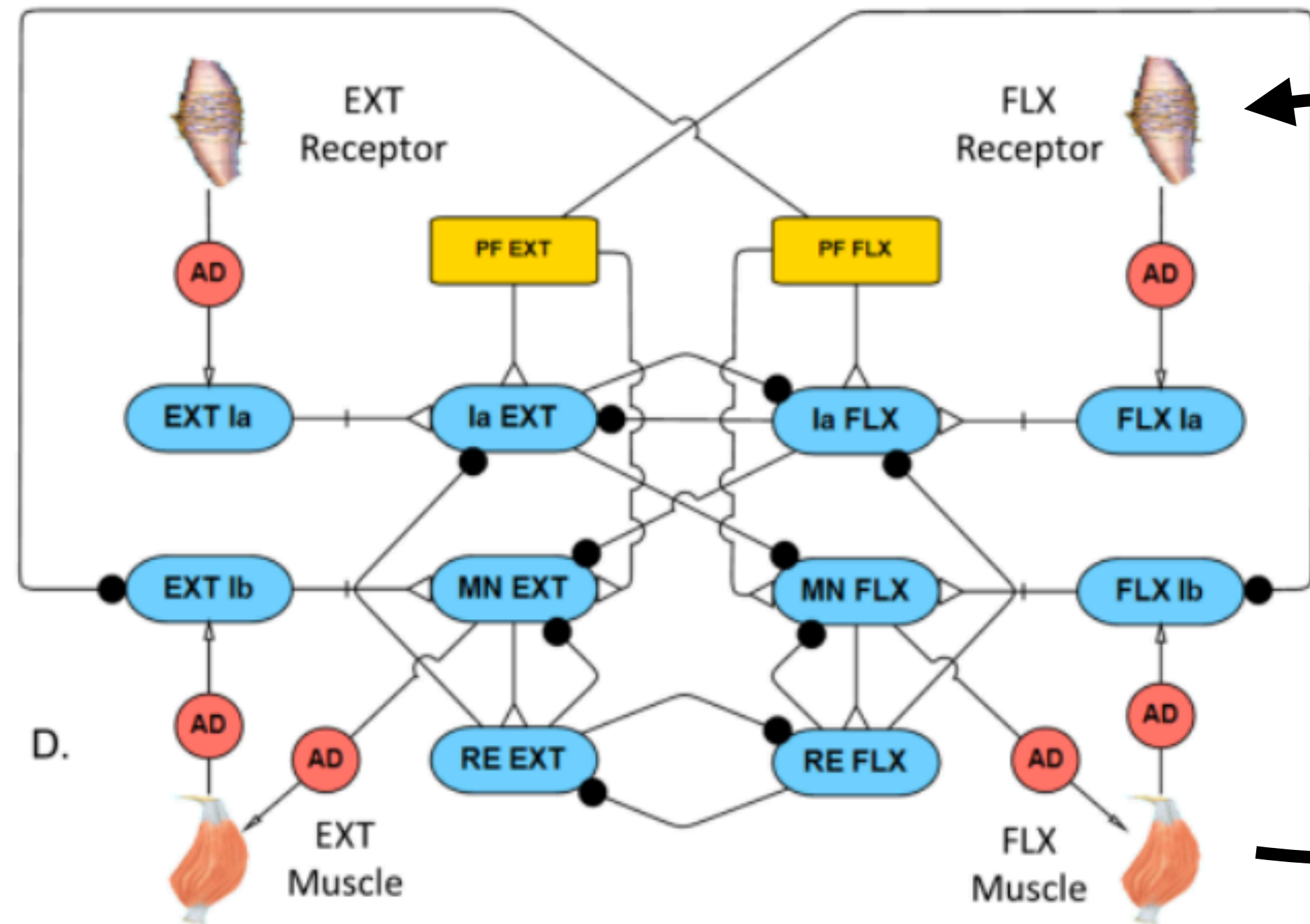
Cheetah Cub (A. T. Spröwitz et al. 2014)



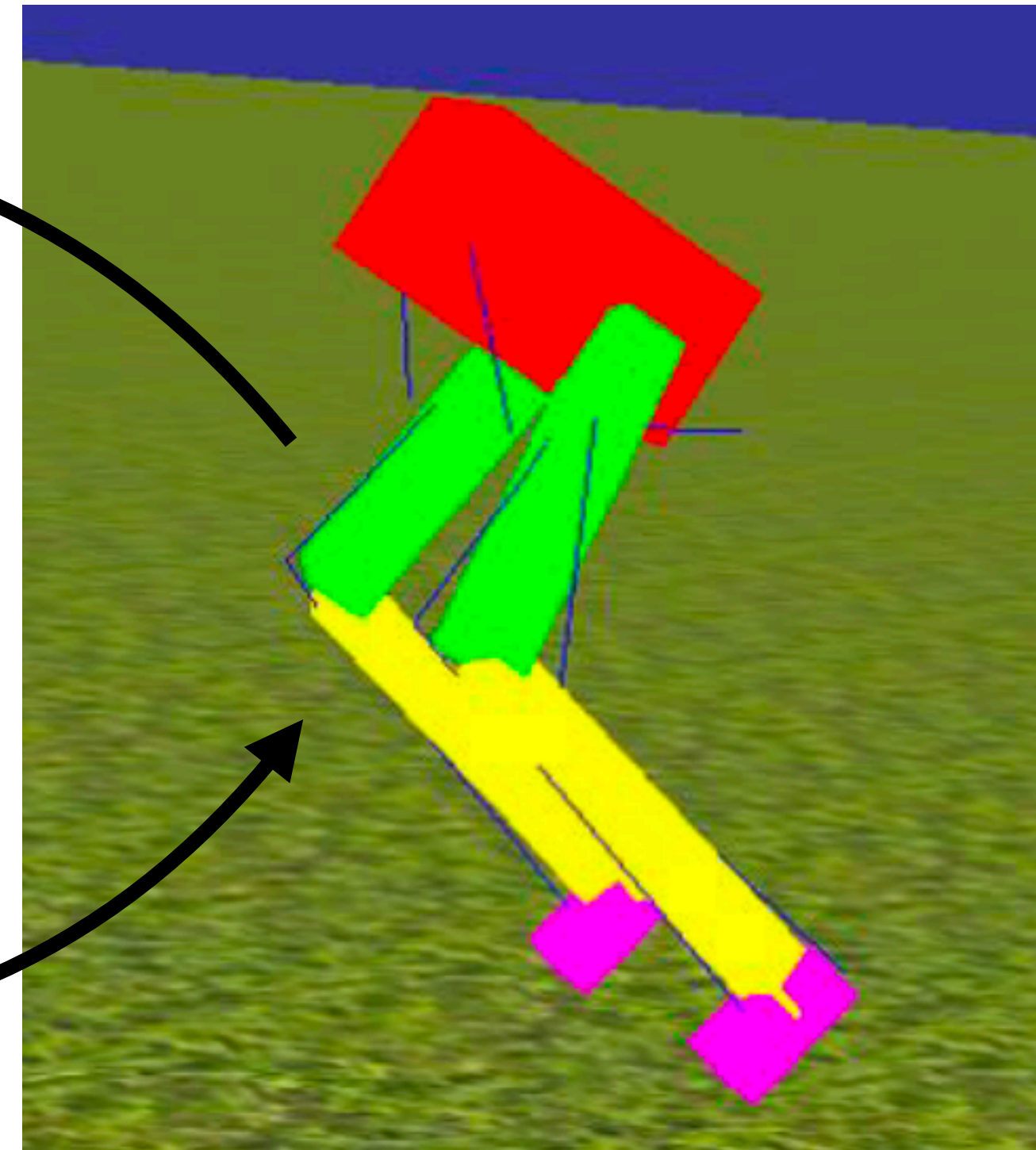
BirdBot (Badri-Spröwitz et al. 2022)



# Synthetic nervous systems (SNS) are a biomimetic form of control



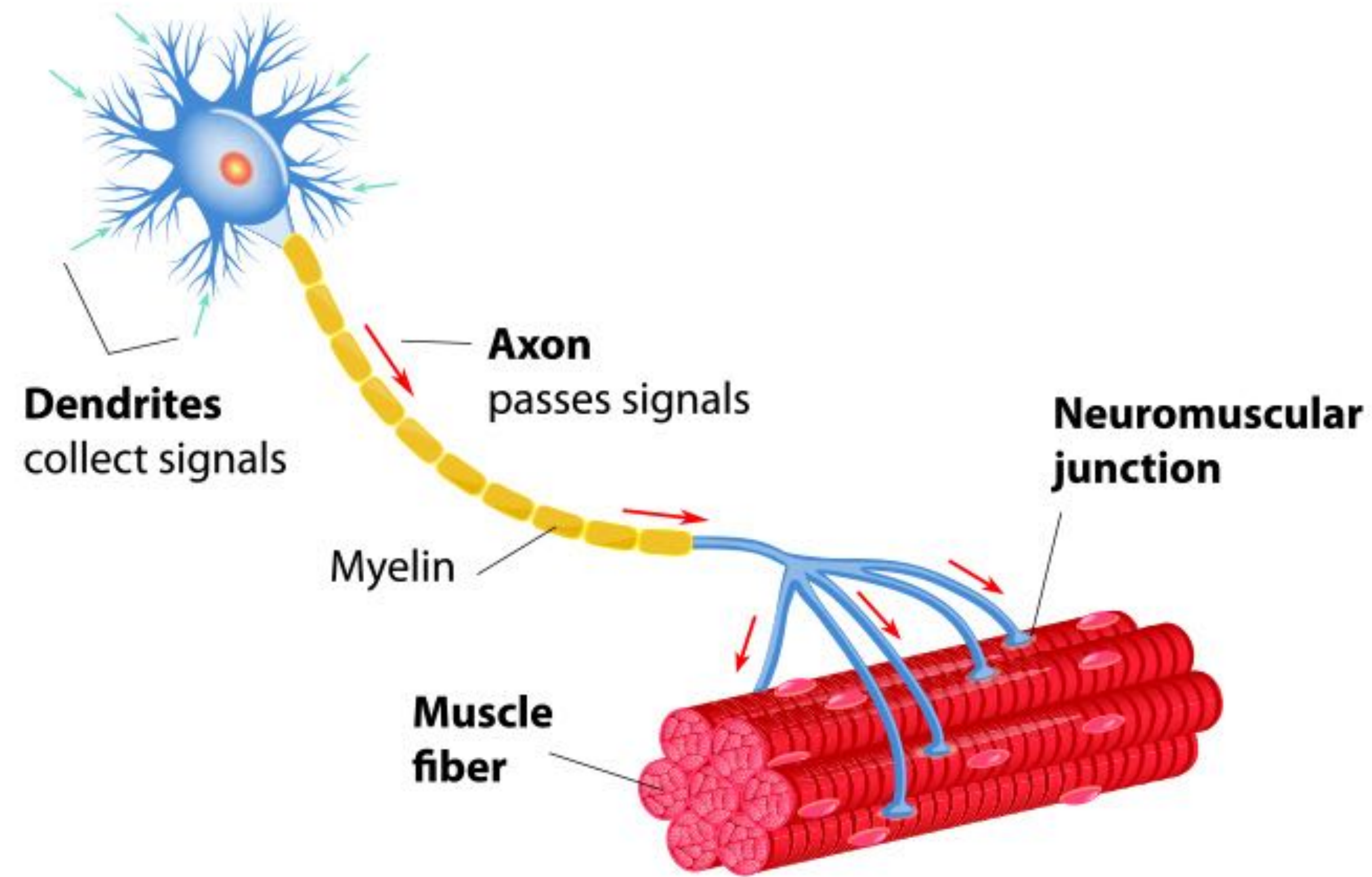
(Deng et al. 2019)



(Deng et al. 2019)

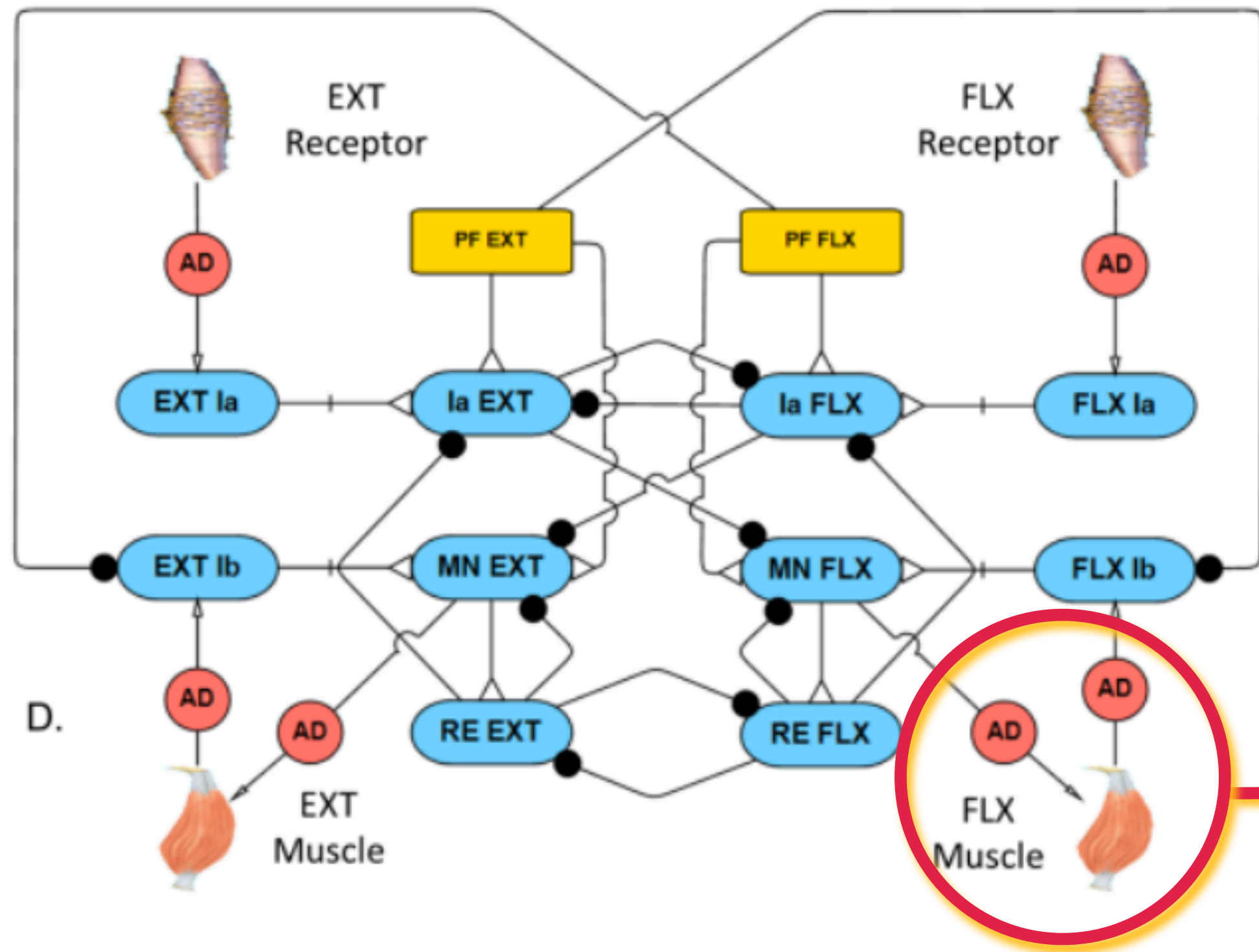


# In many biological systems, muscles are activated in *pulses* via *spiking* motoneurons

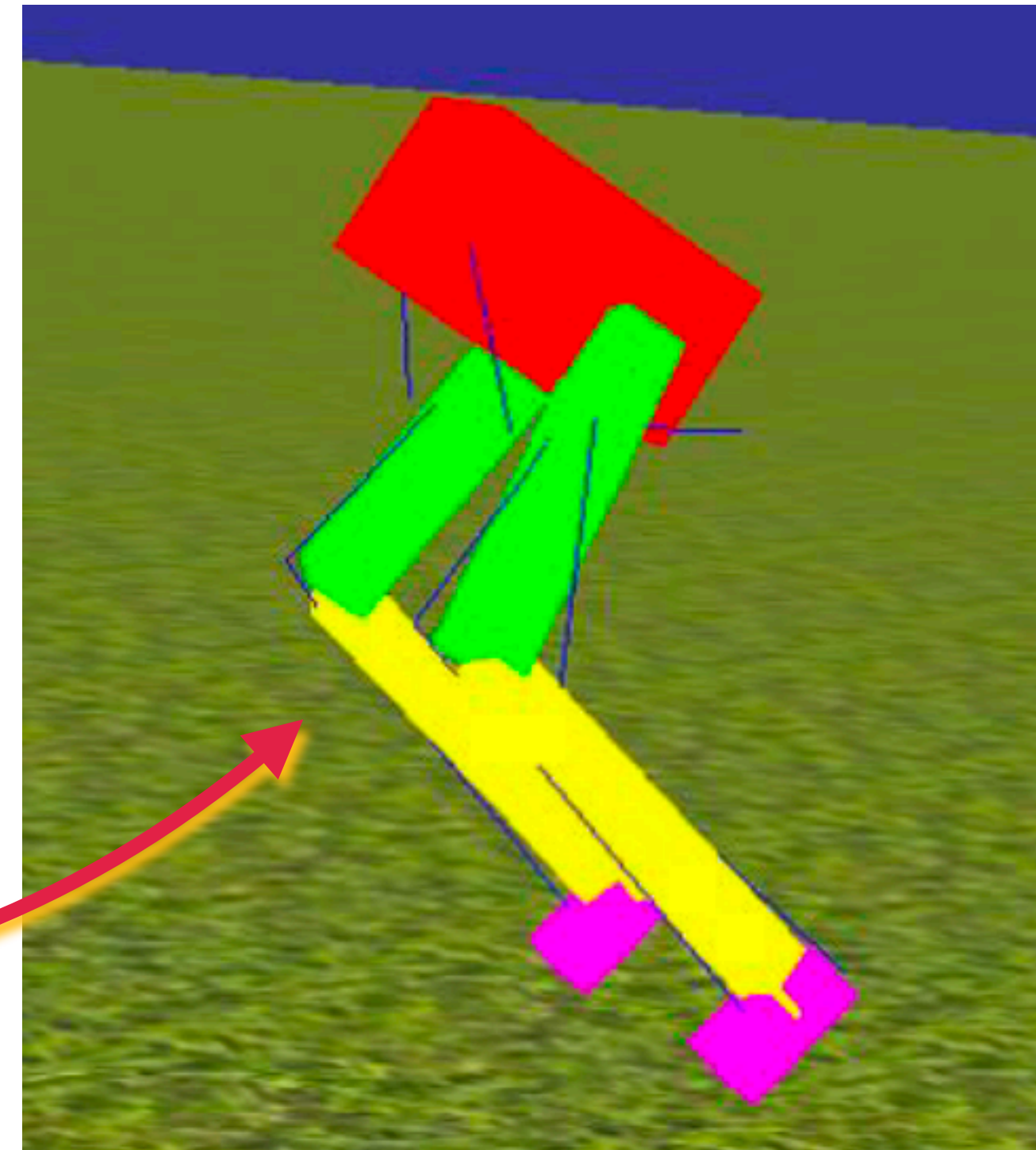




However, many walking models utilize *non-spiking* motoneuron activations



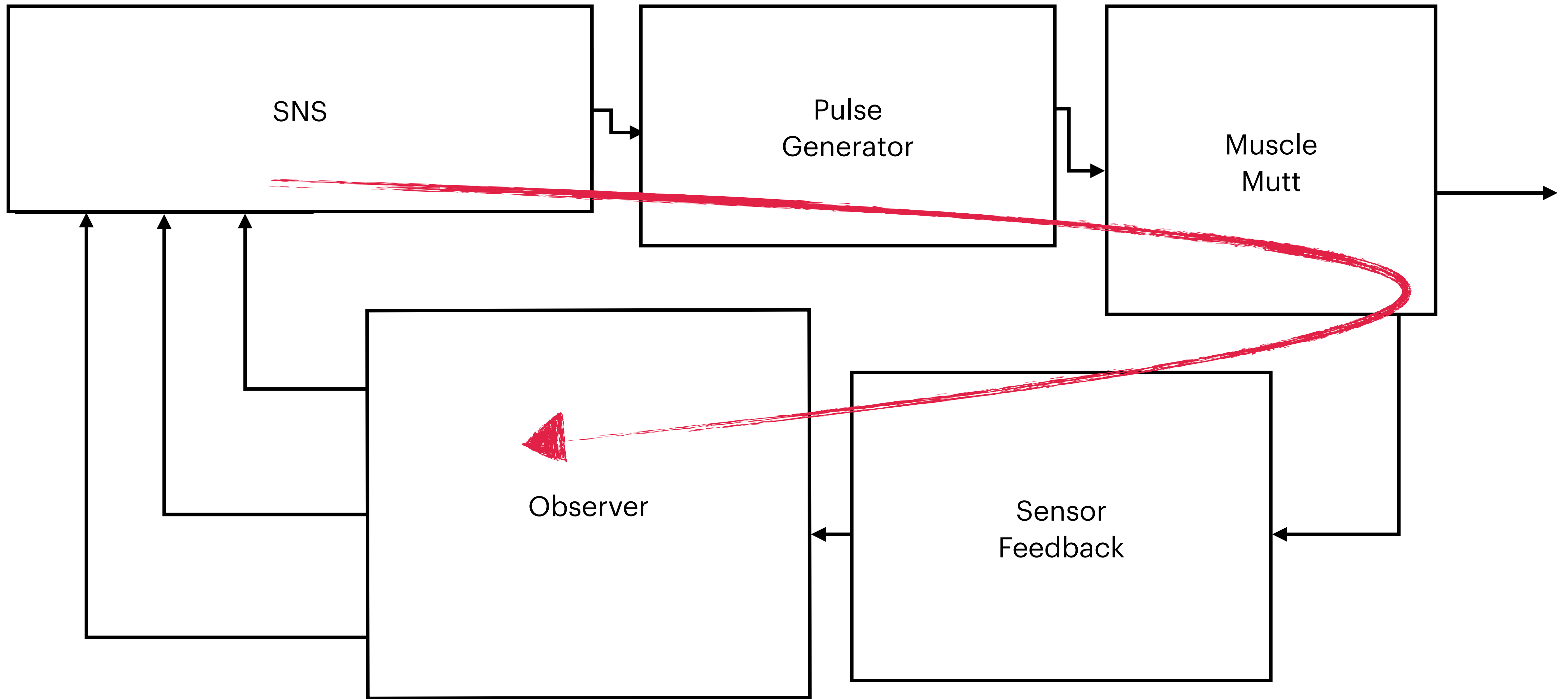
(Deng et al. 2019)



(Deng et al. 2019)

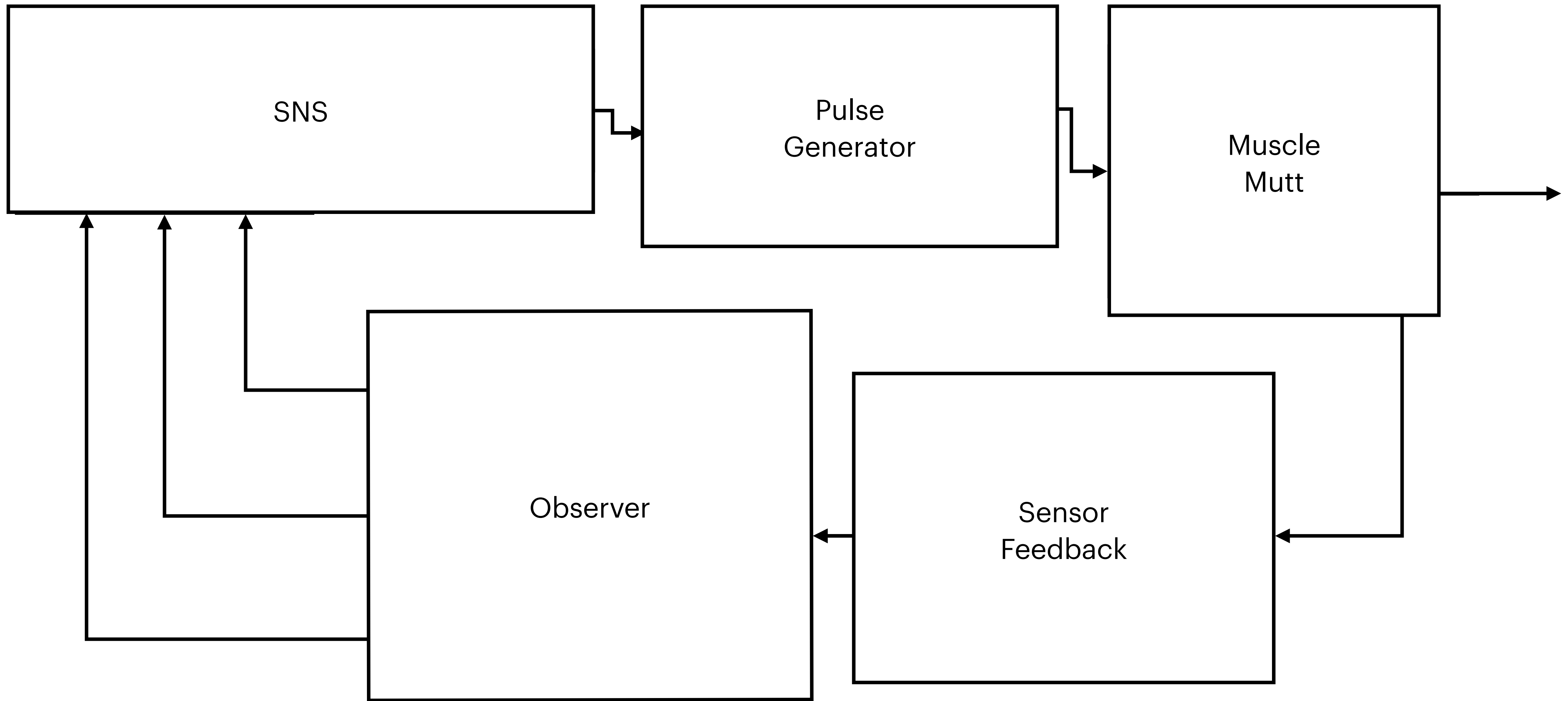


# What are we talking about?



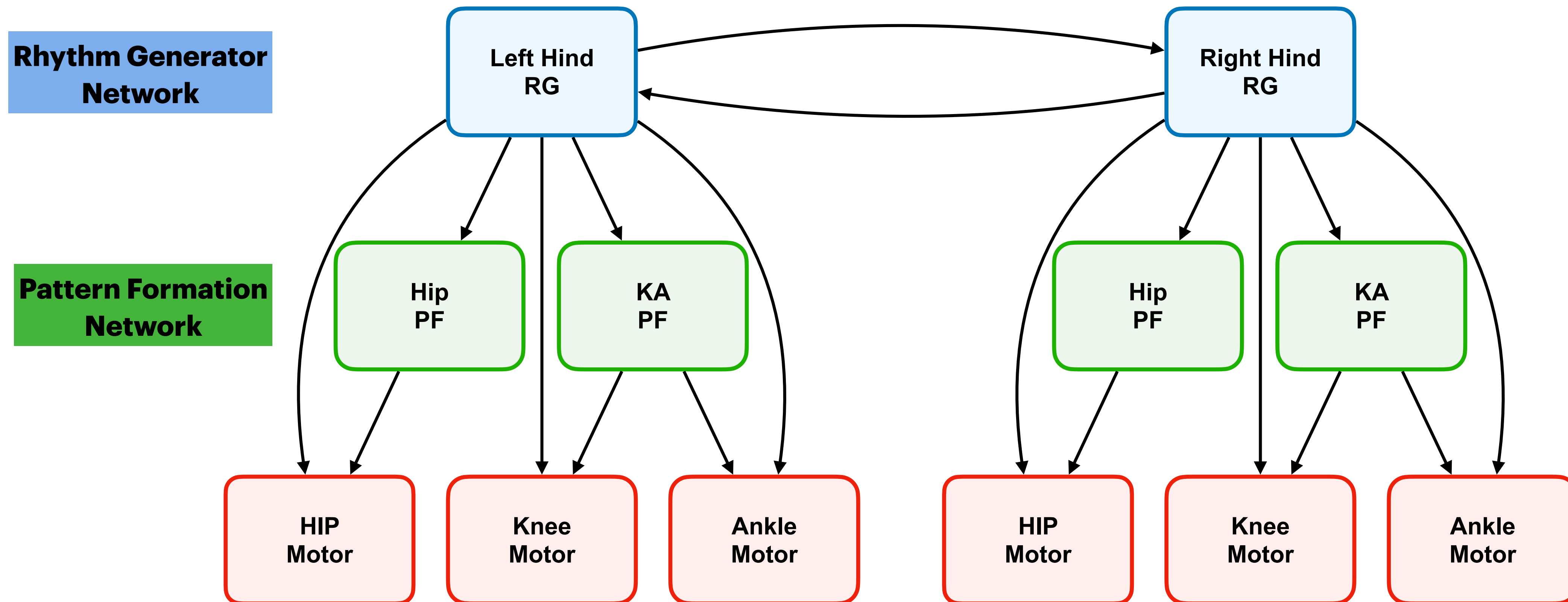


# SNS controls Muscle Mutt locomotion

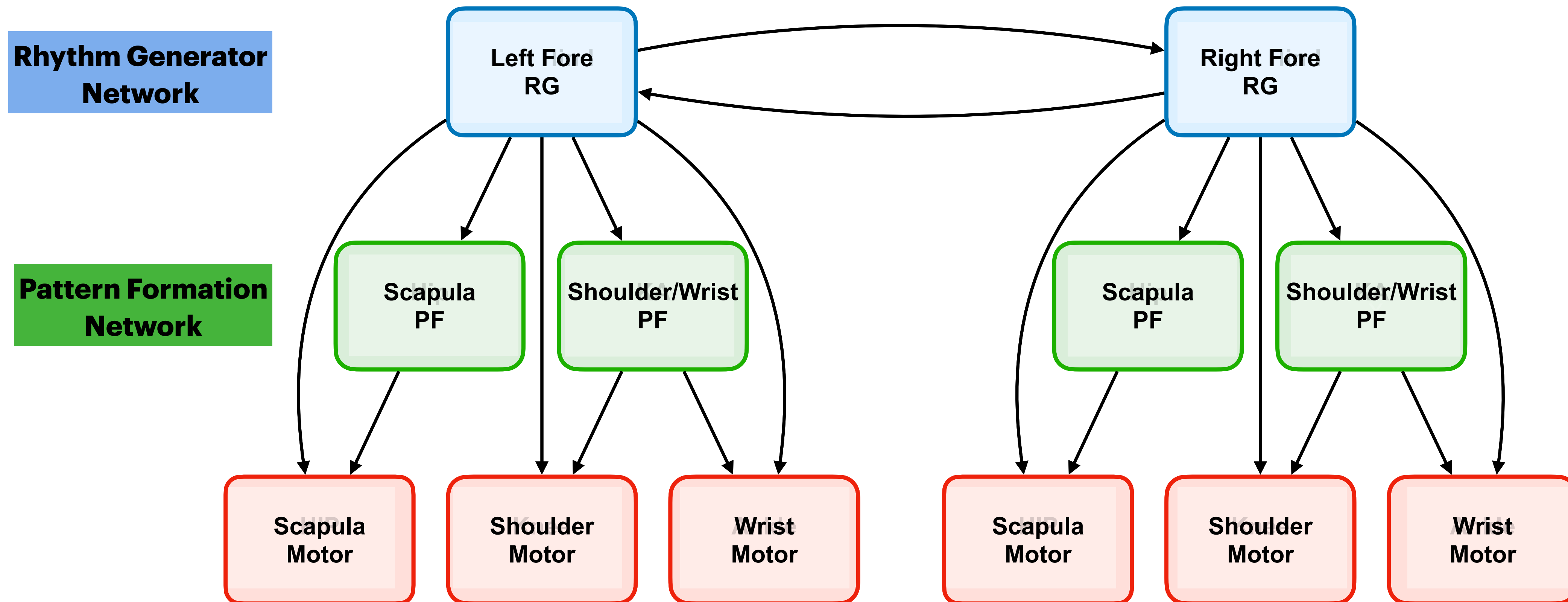




# Rat hind limb SNS controls Mutt's hind limbs

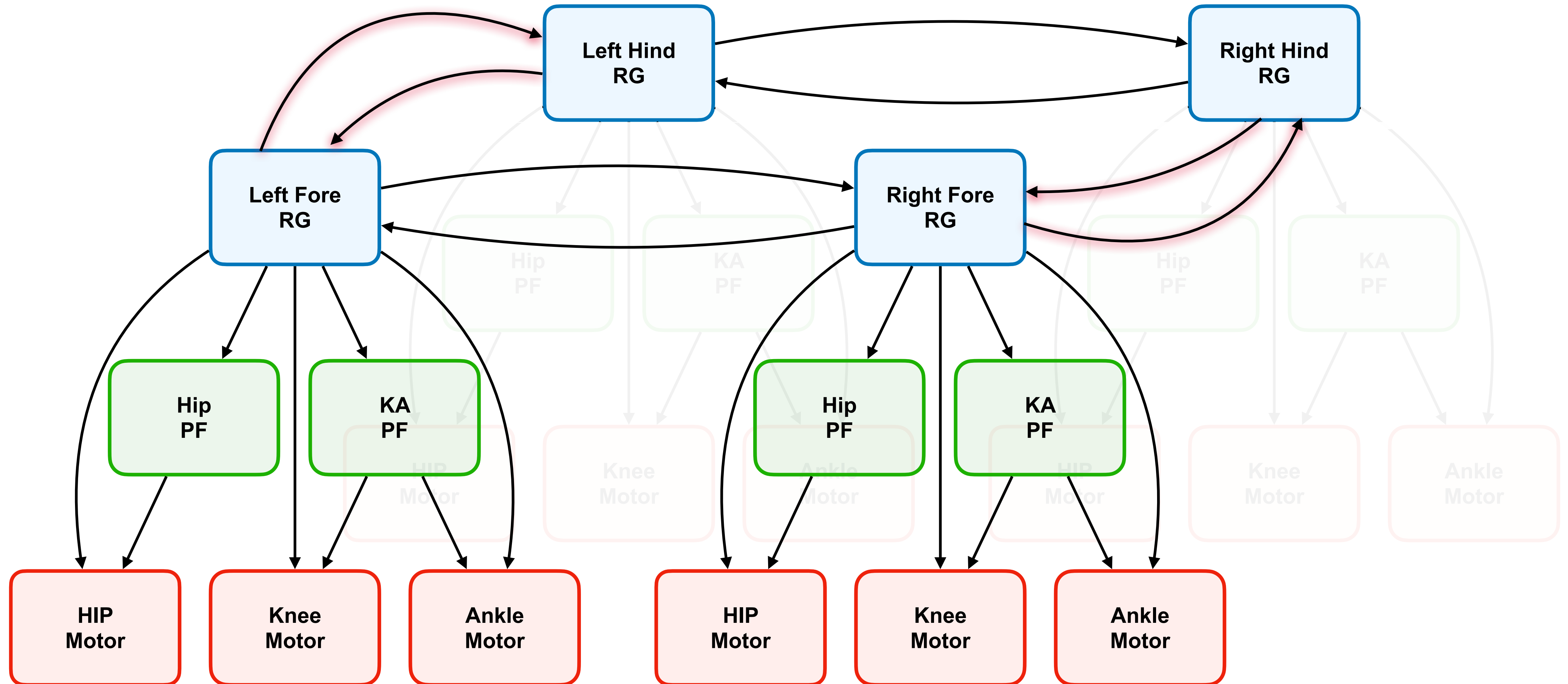


# Fore limb SNS is a copy of hind limb SNS

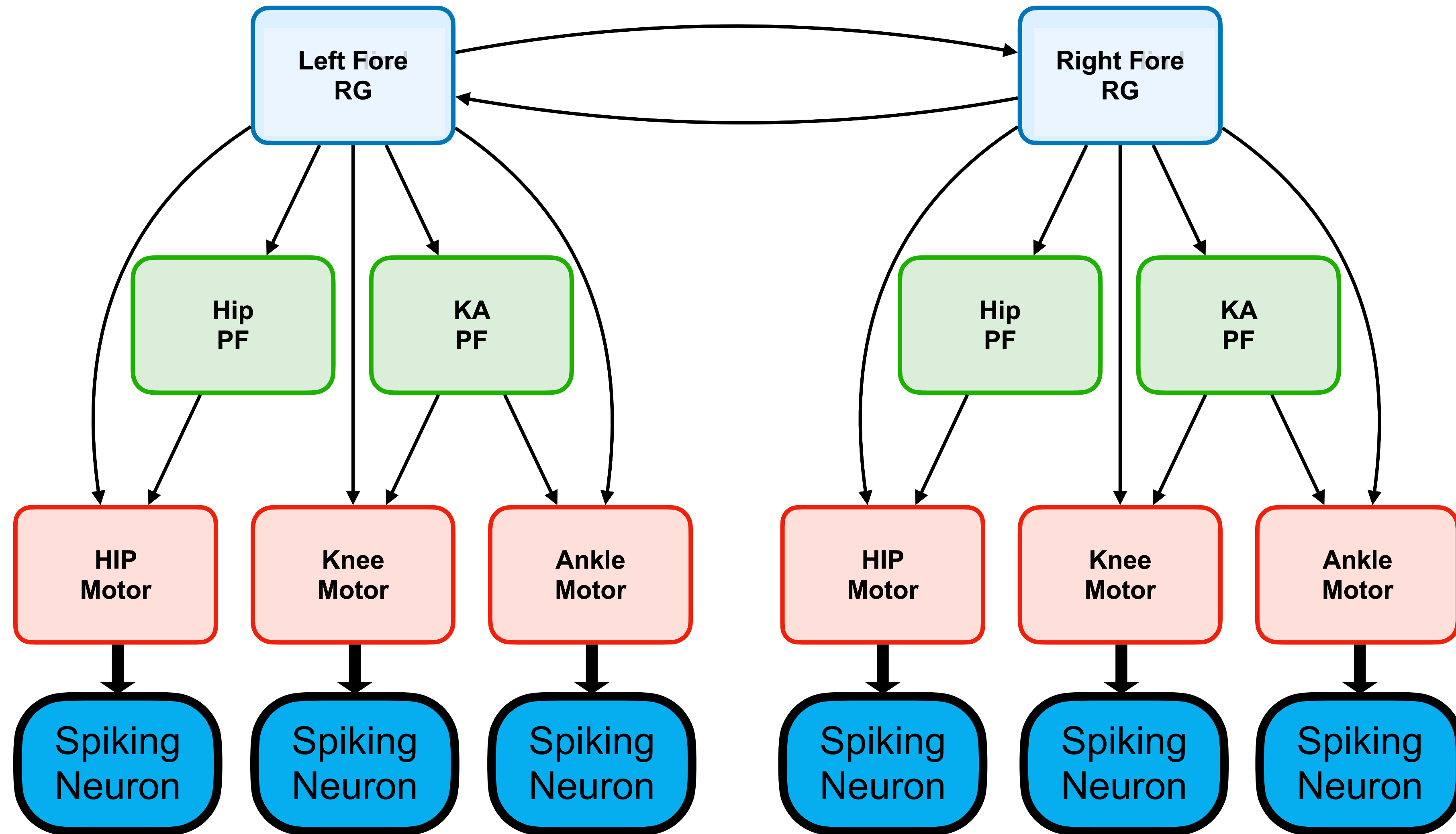




# ***New fore/hind limb connections are based on left/right connections***

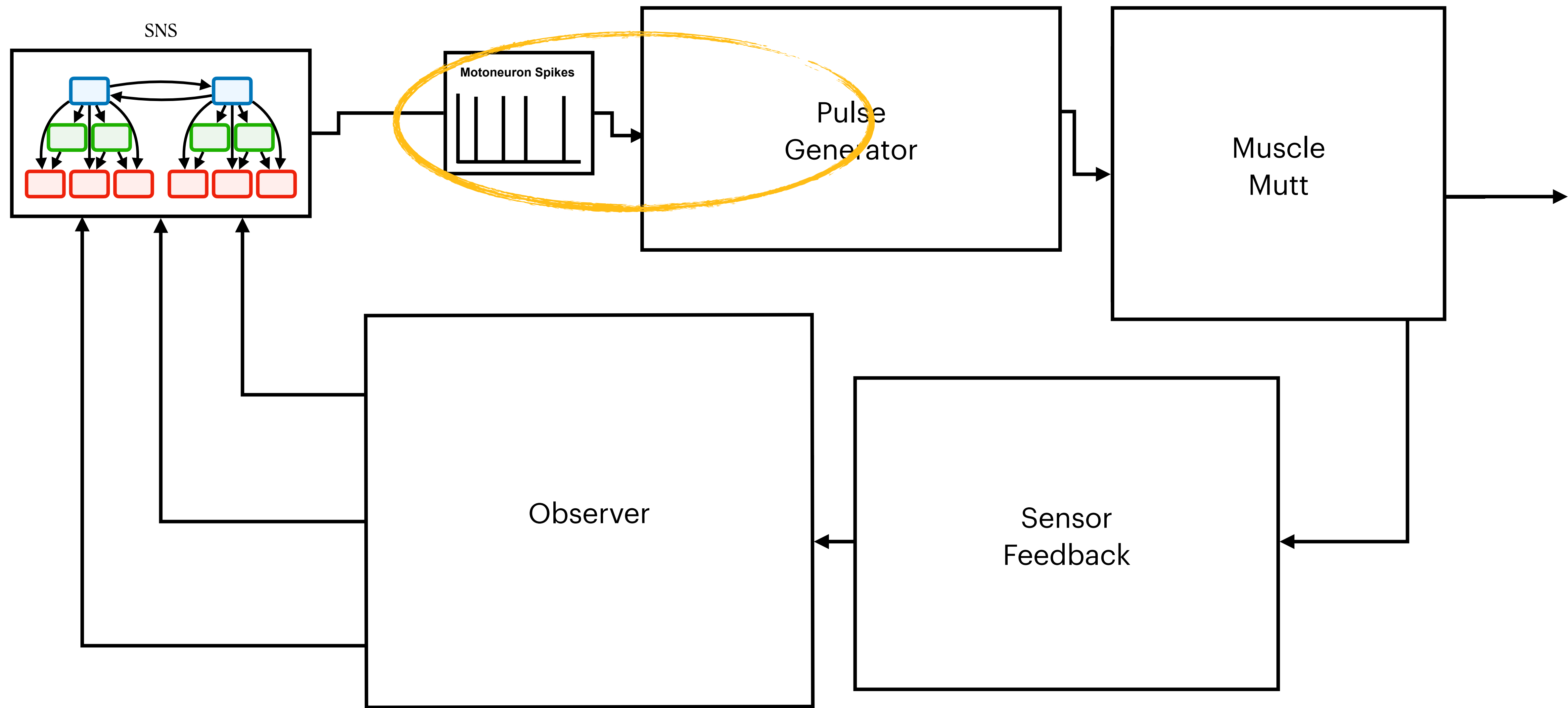


# *New spiking motoneurons are connected to existing non-spiking neurons*

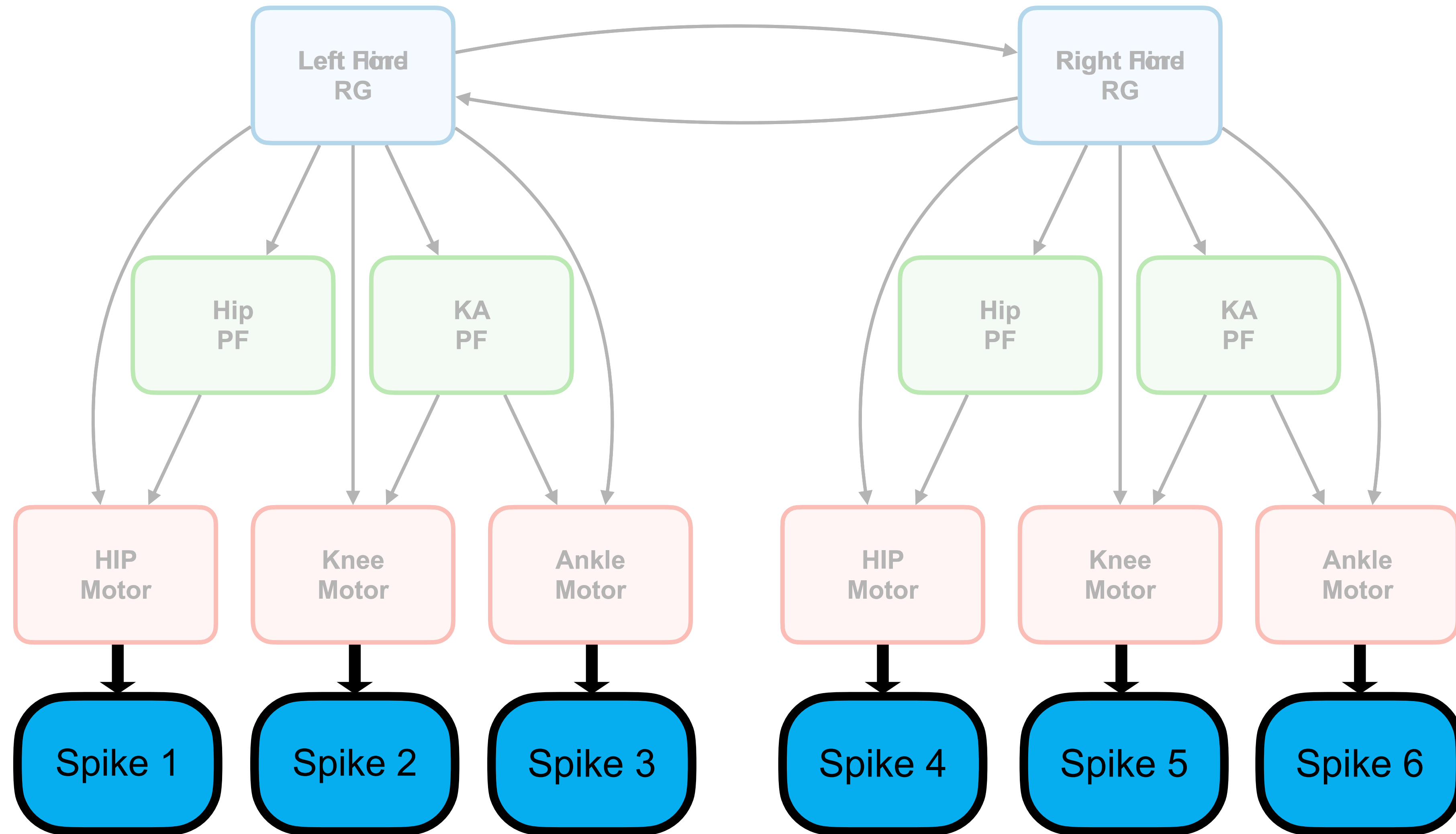




# Communication between PC and microcontroller is difficult

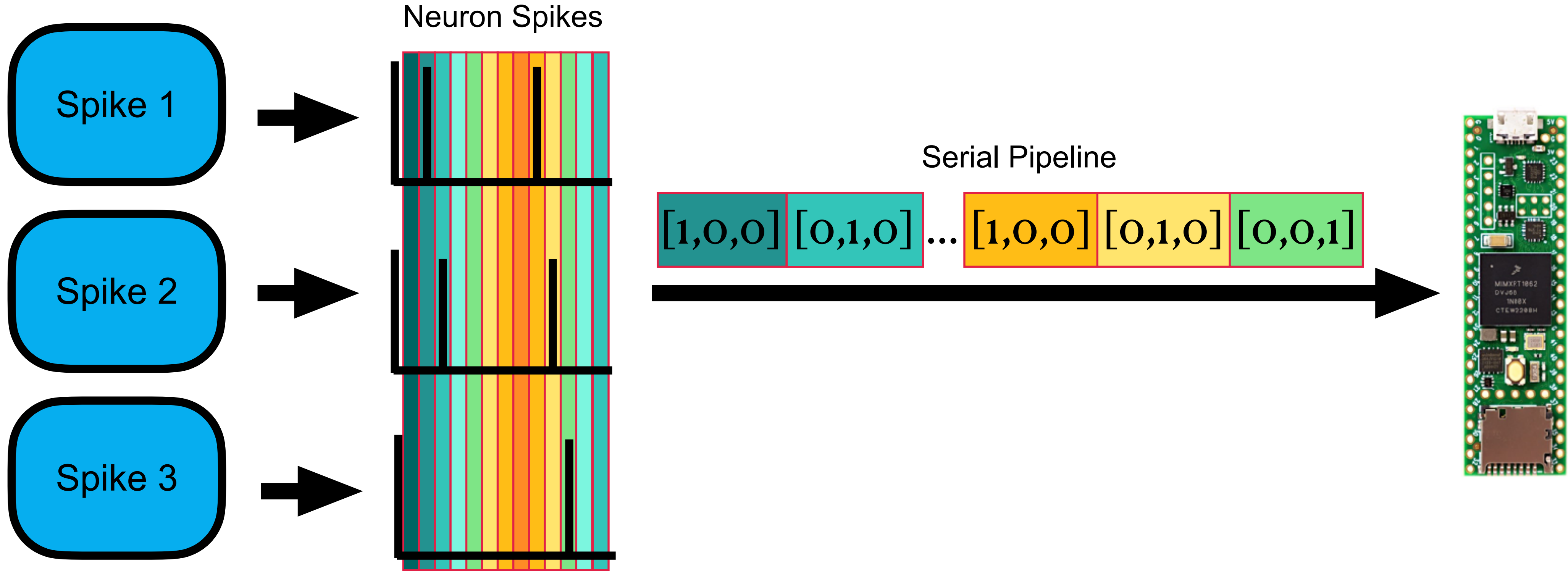


# Looking back at our spiking neurons...

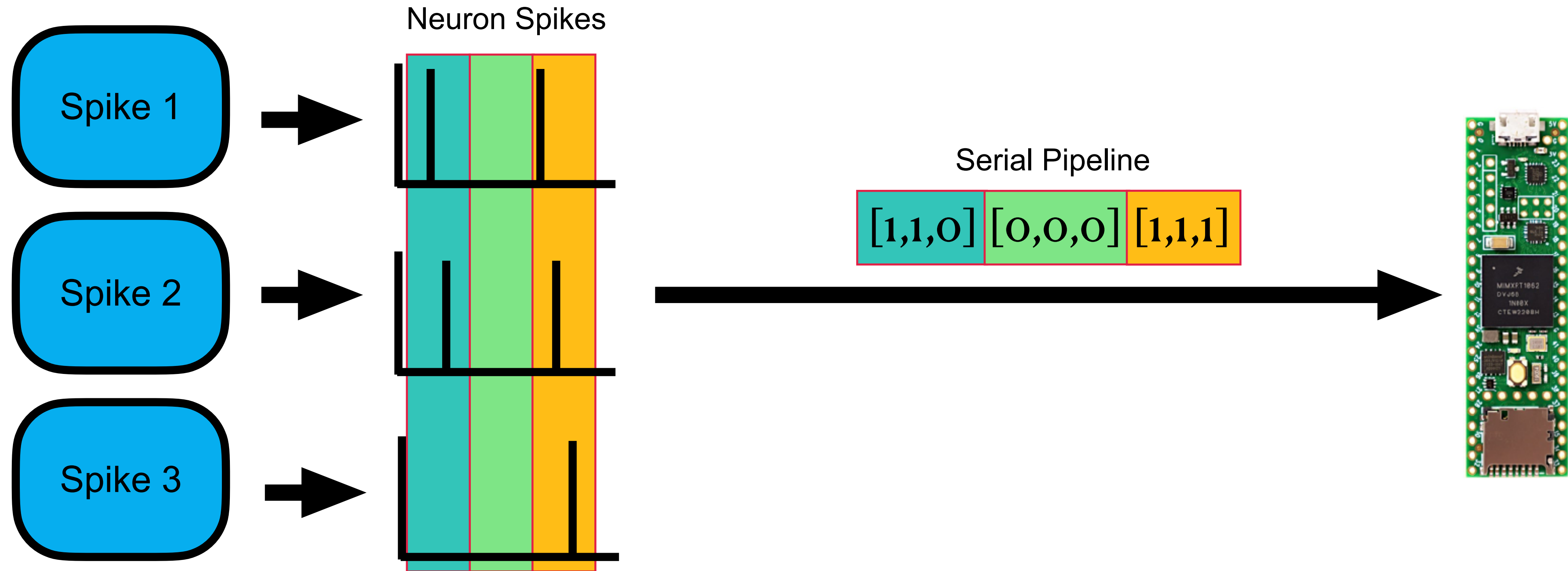




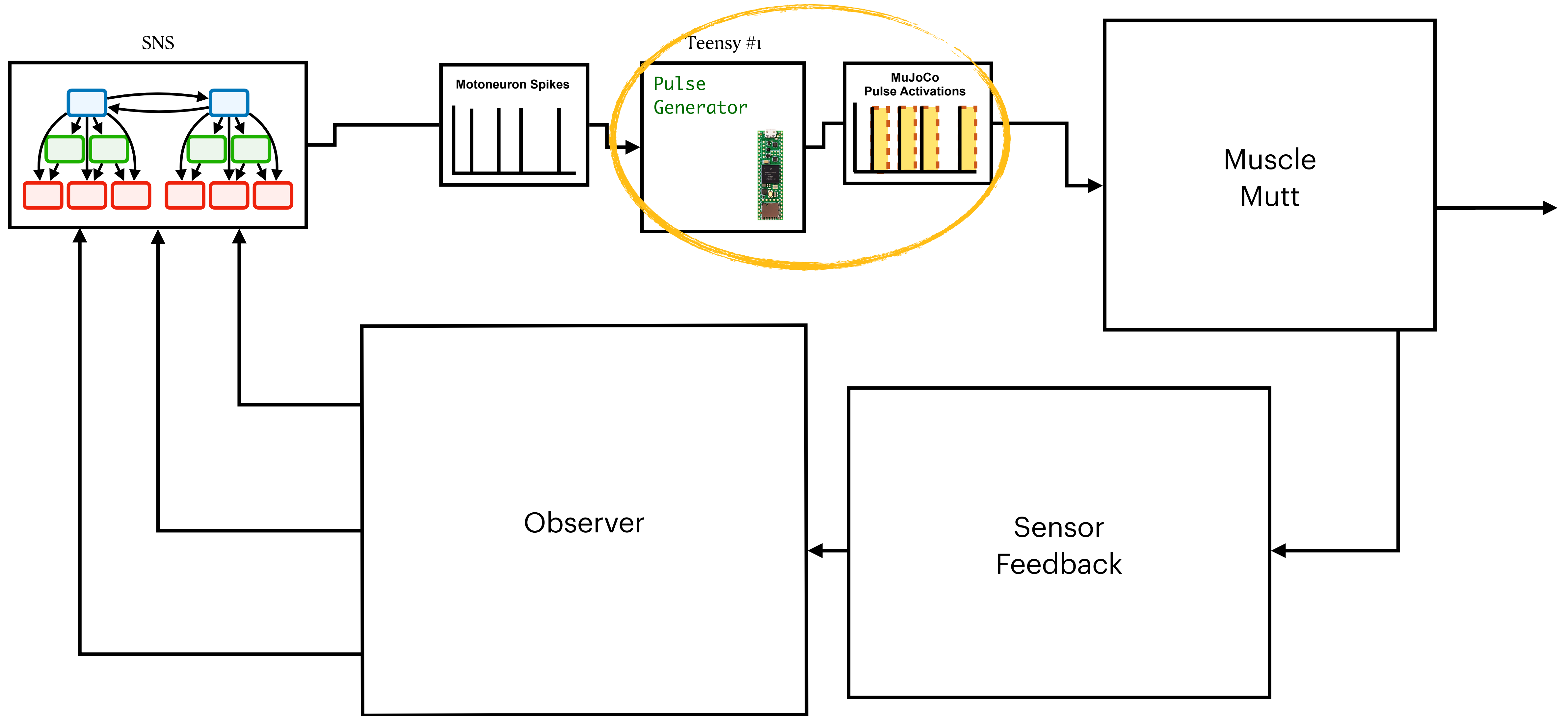
# Neuron spikes create a *serial pipeline bottleneck*.



# Fixed communication frequency removes serial bottleneck.

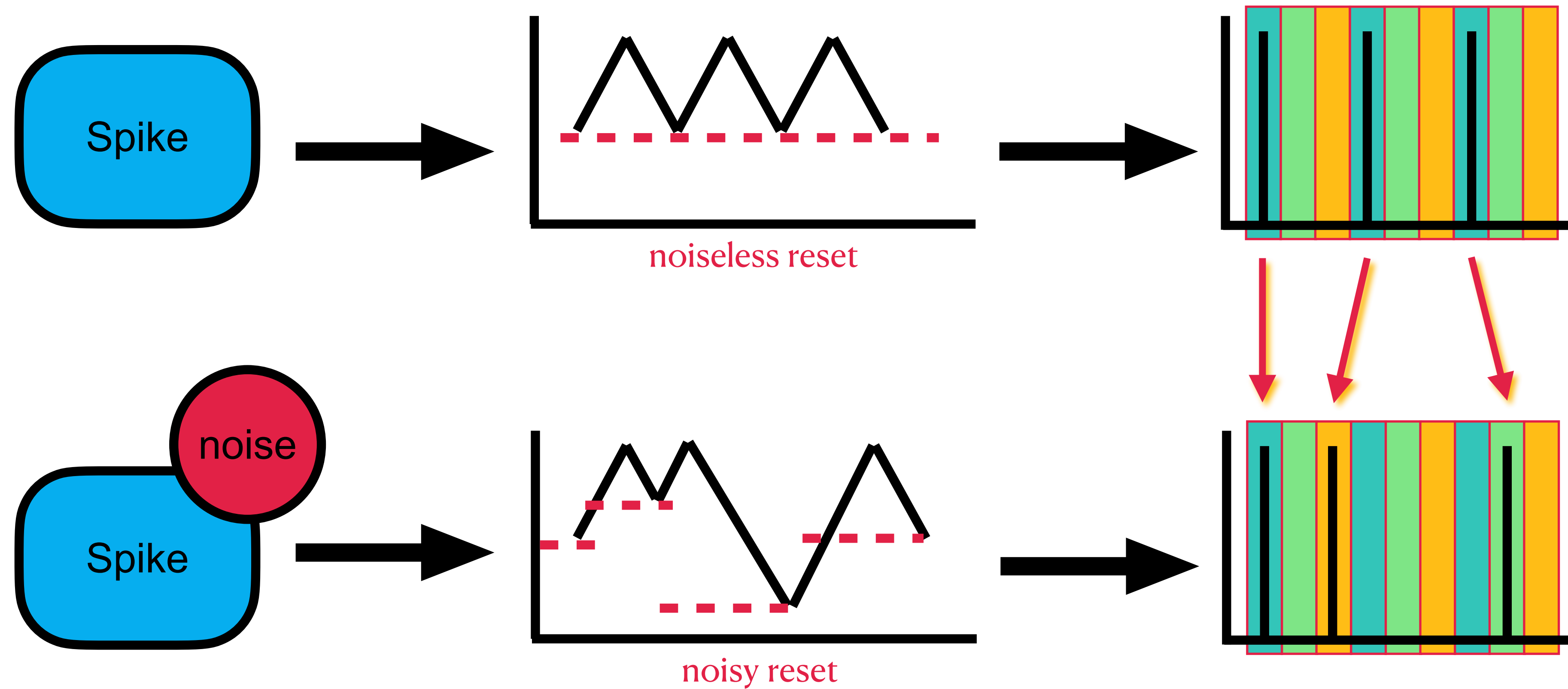


# BPA's can be activated via neuron spikes

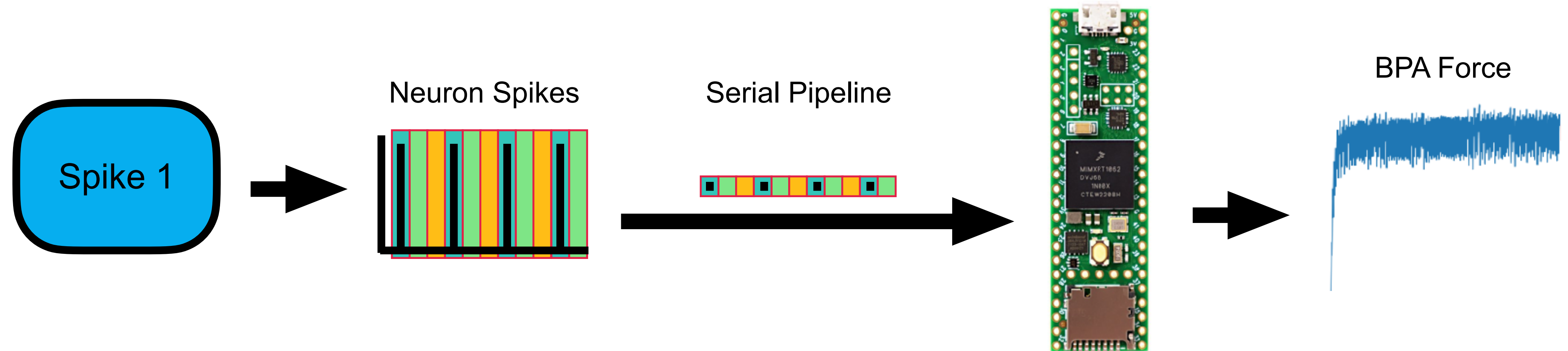




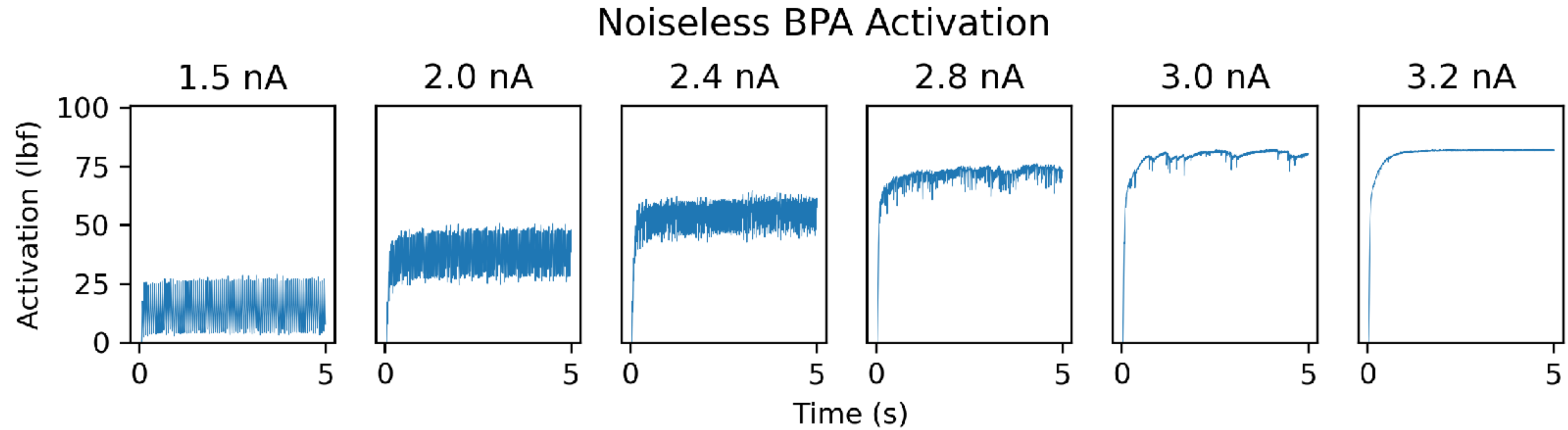
# I edited the SNS-Toolbox backend to create a noisy reset neuron



# Neuron spikes can be used to generate pulse-based activations in BPAs

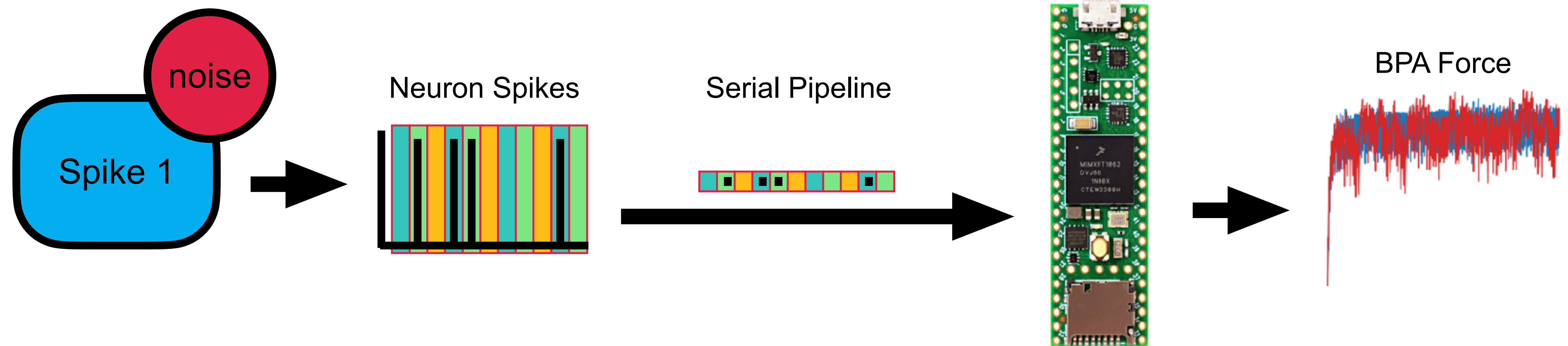


# Spike rate increases BPA mean force output until saturation

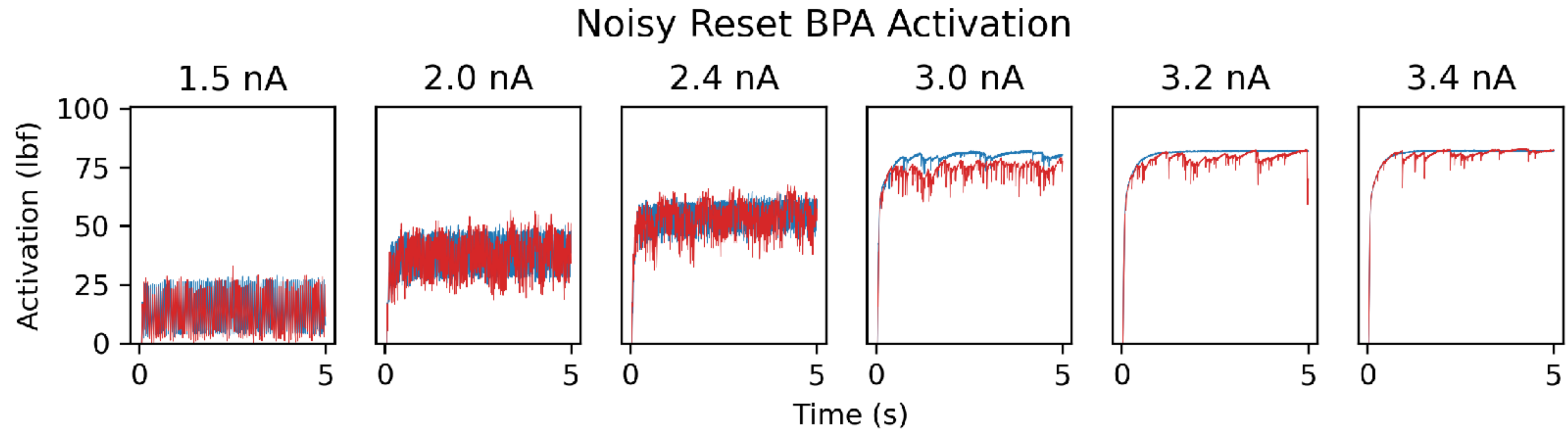




# Noisy spiking parameter modulates the BPA mean force

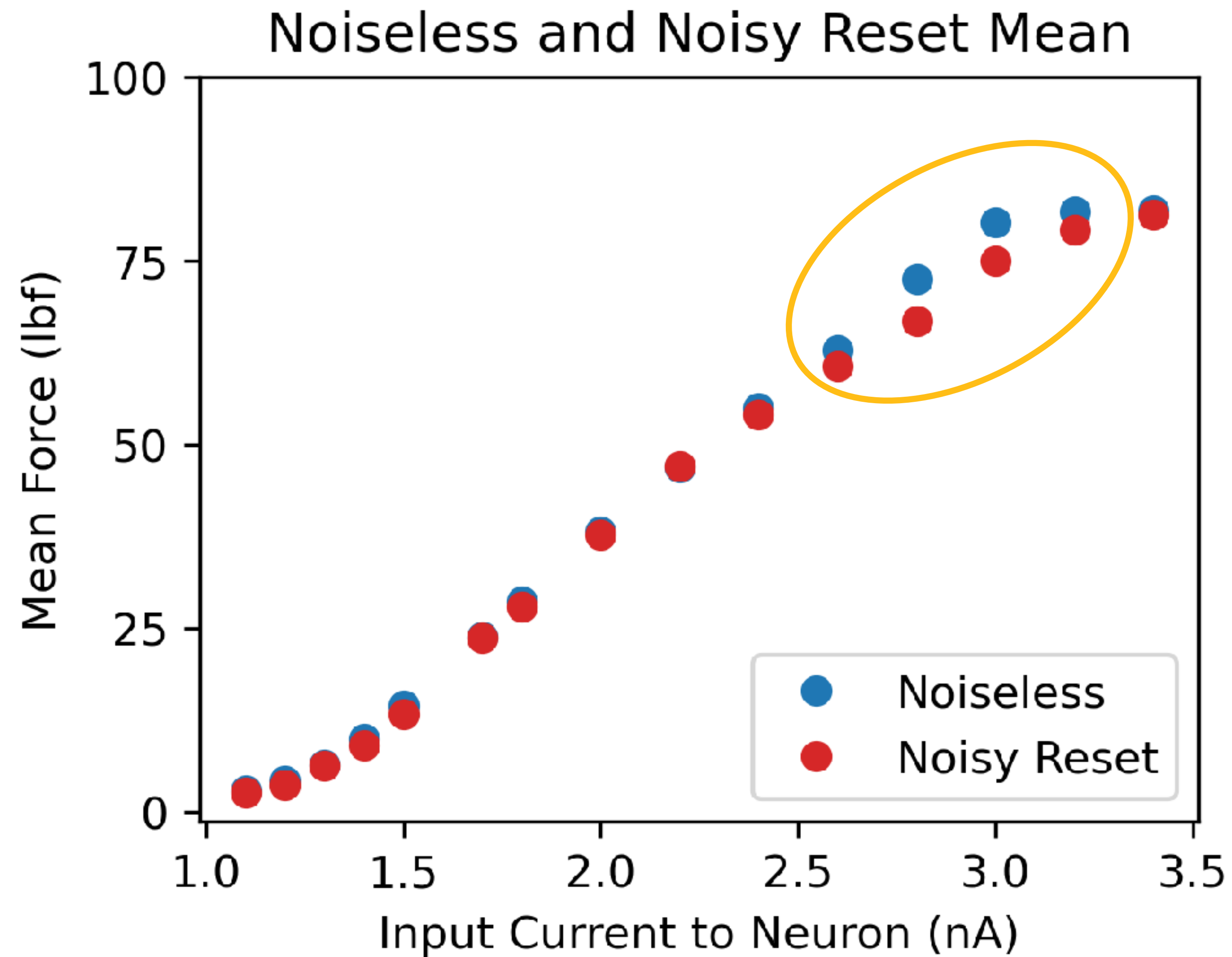


# Noisy spikes decrease the mean force at high frequency spikes



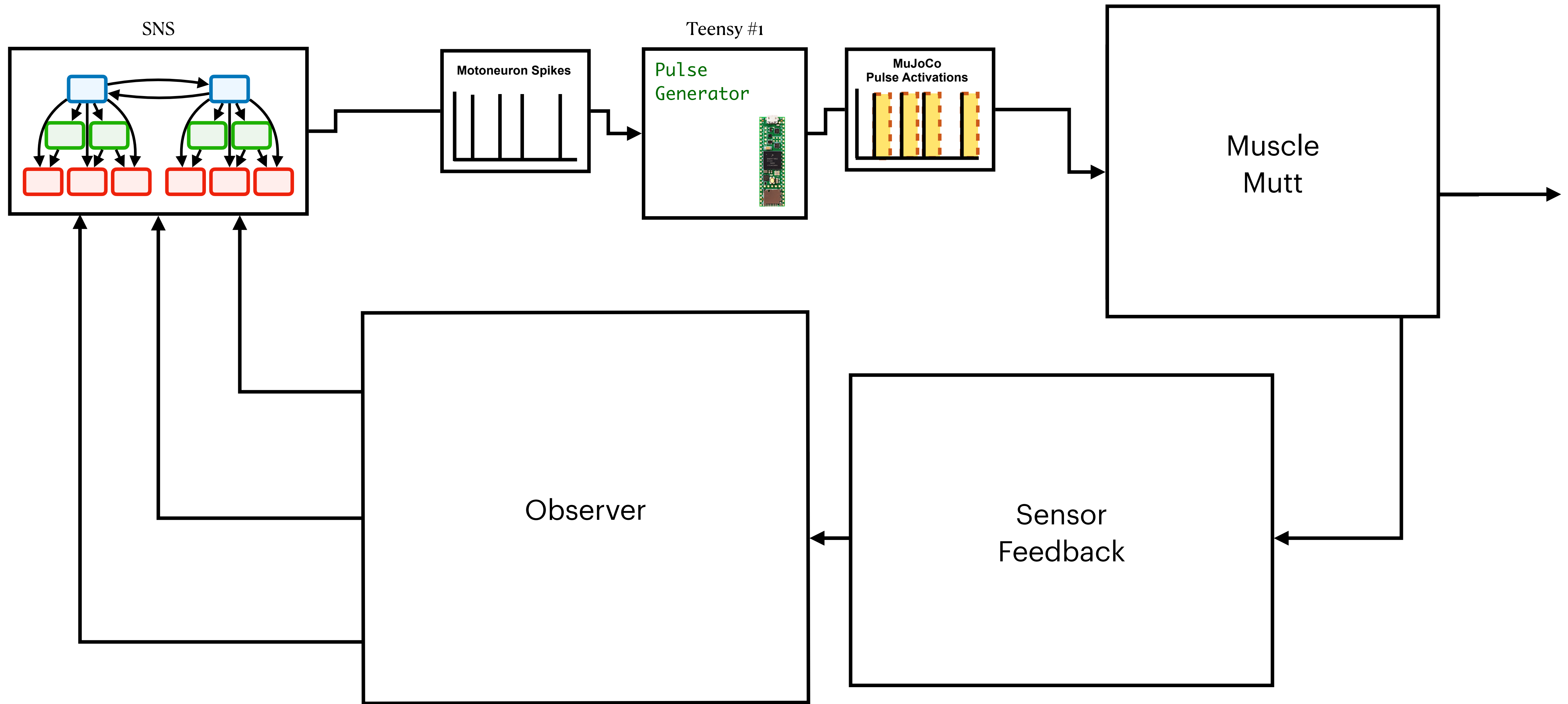
Mean force outputs from **noisy** (red) and **noiseless** (blue) neurons given different input currents.

# Noisy pulses can modify the mean activation force up to the saturation frequency

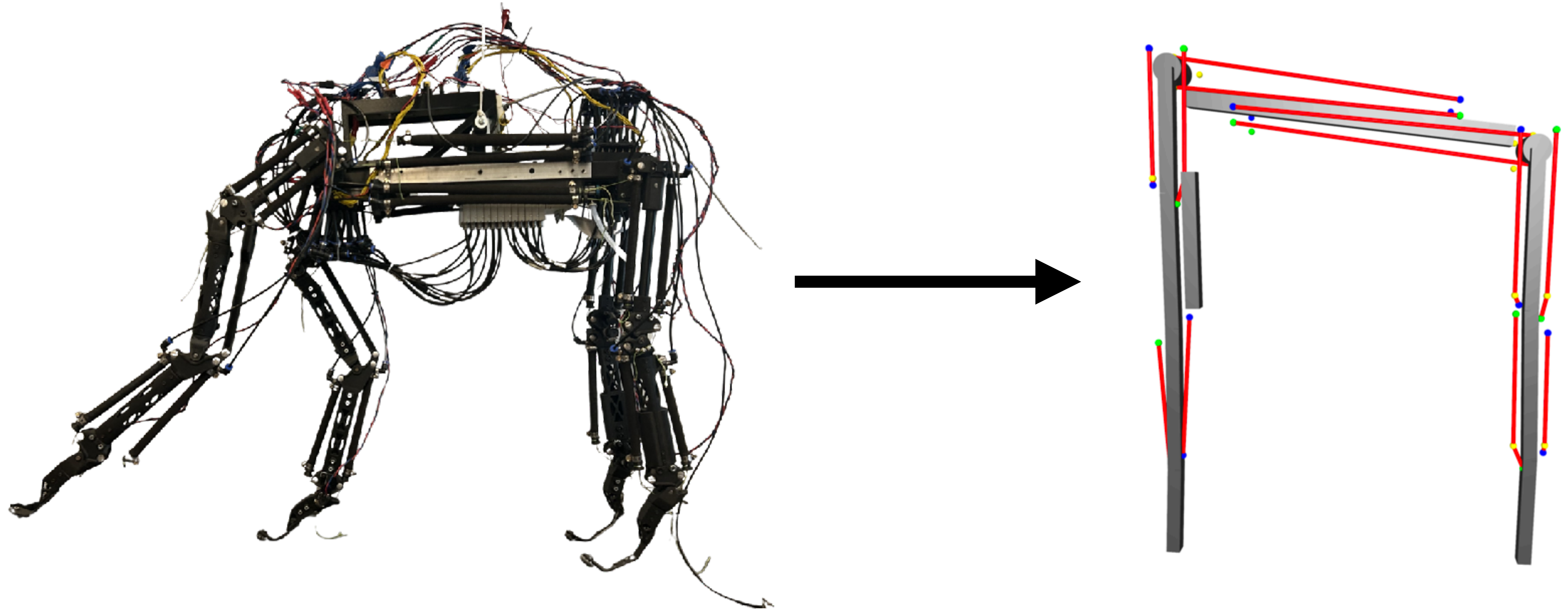




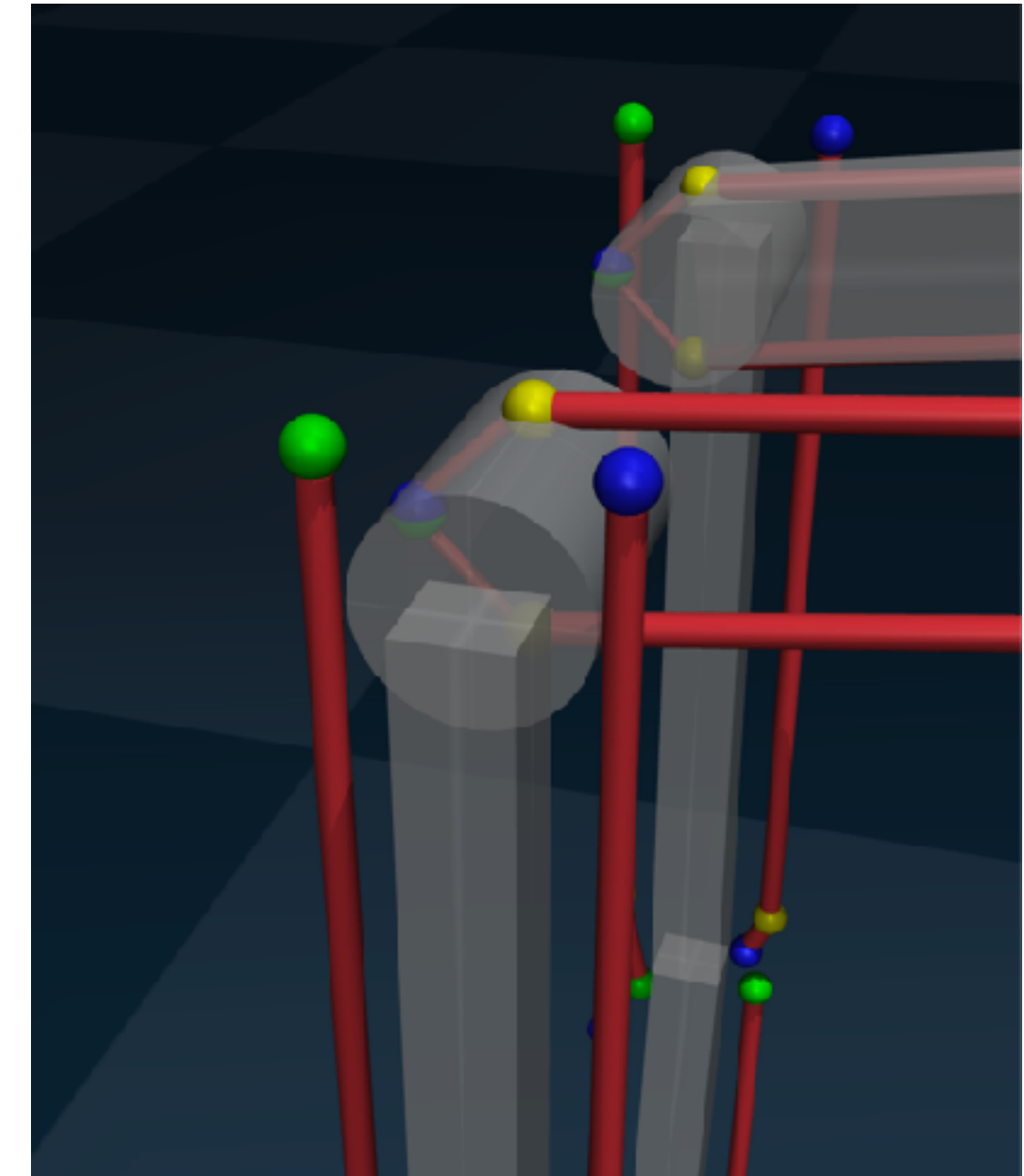
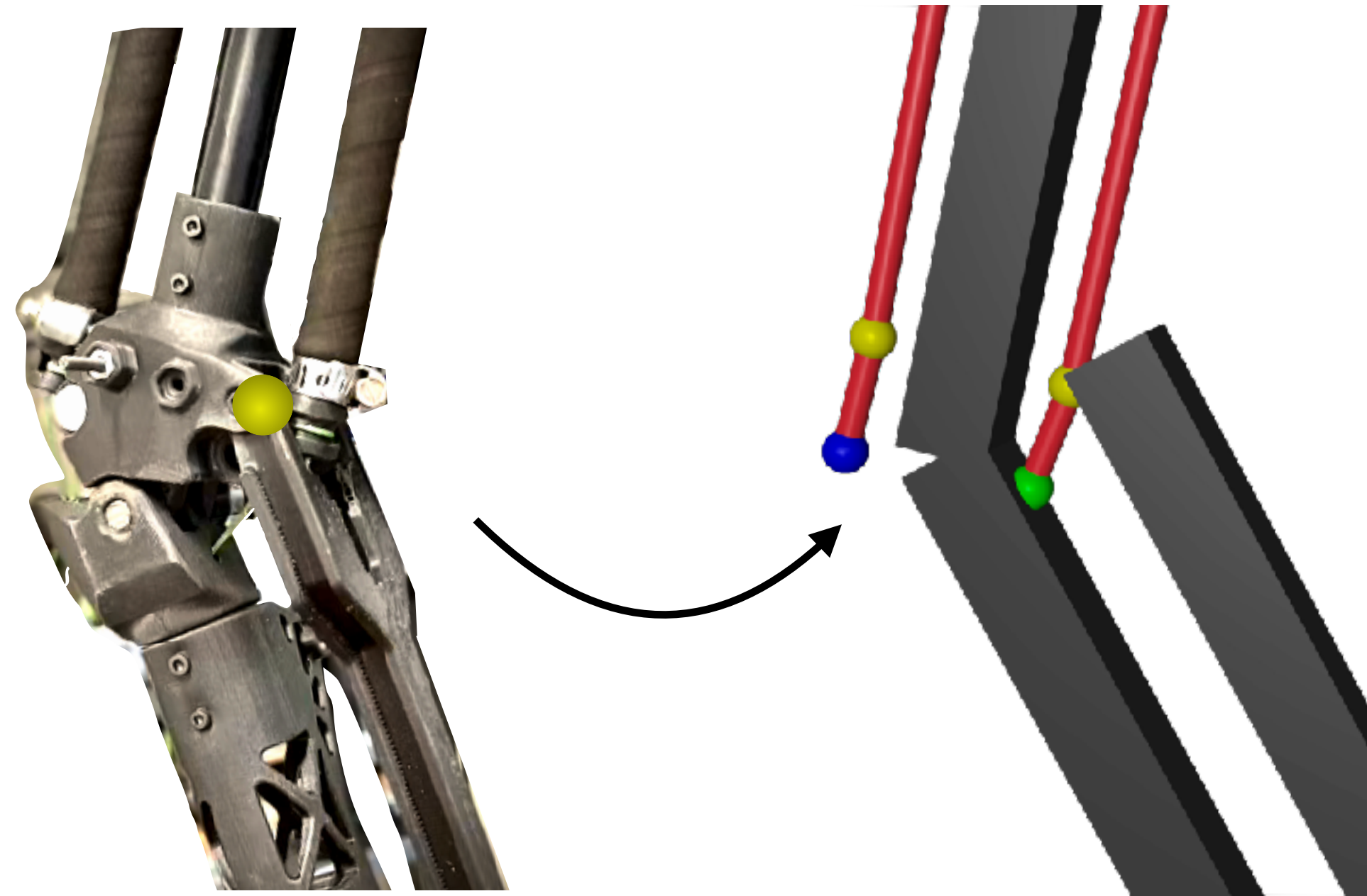
# The SNS must be tested before implementation on Muscle Mutt



# A MuJoCo model is used to test SNS effectiveness

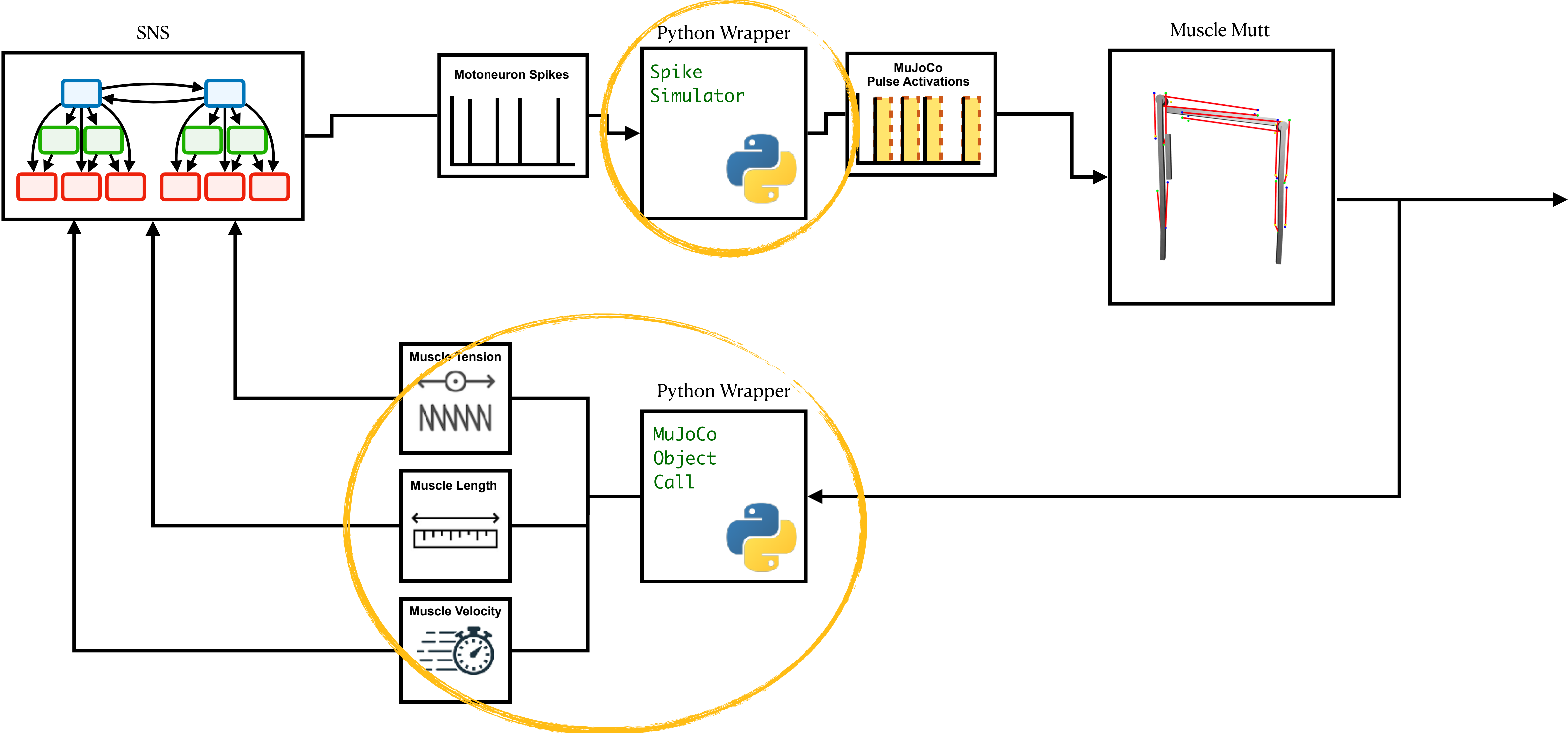


# MuJoCo muscles are modeled as linear actuators with wrap points

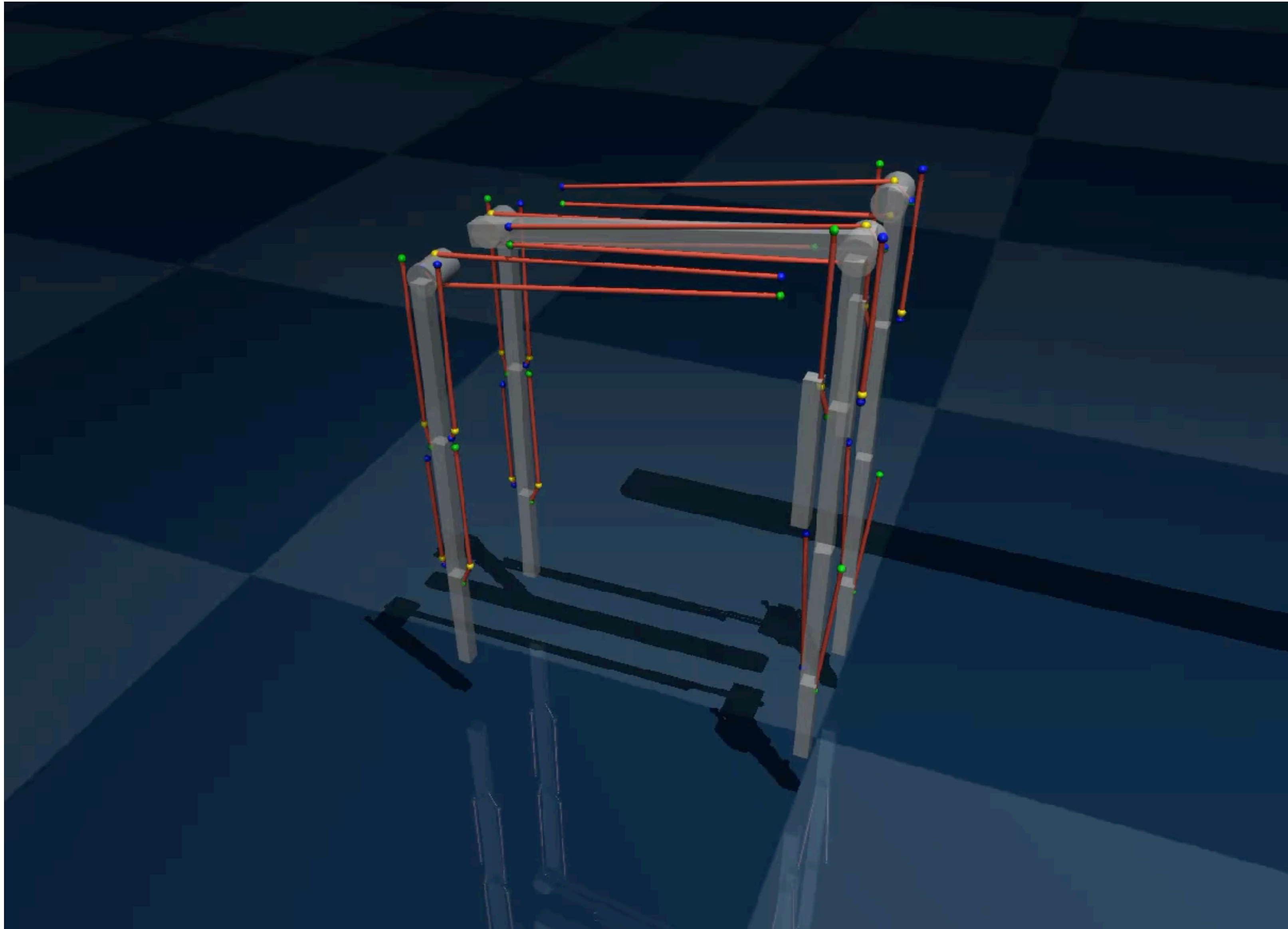




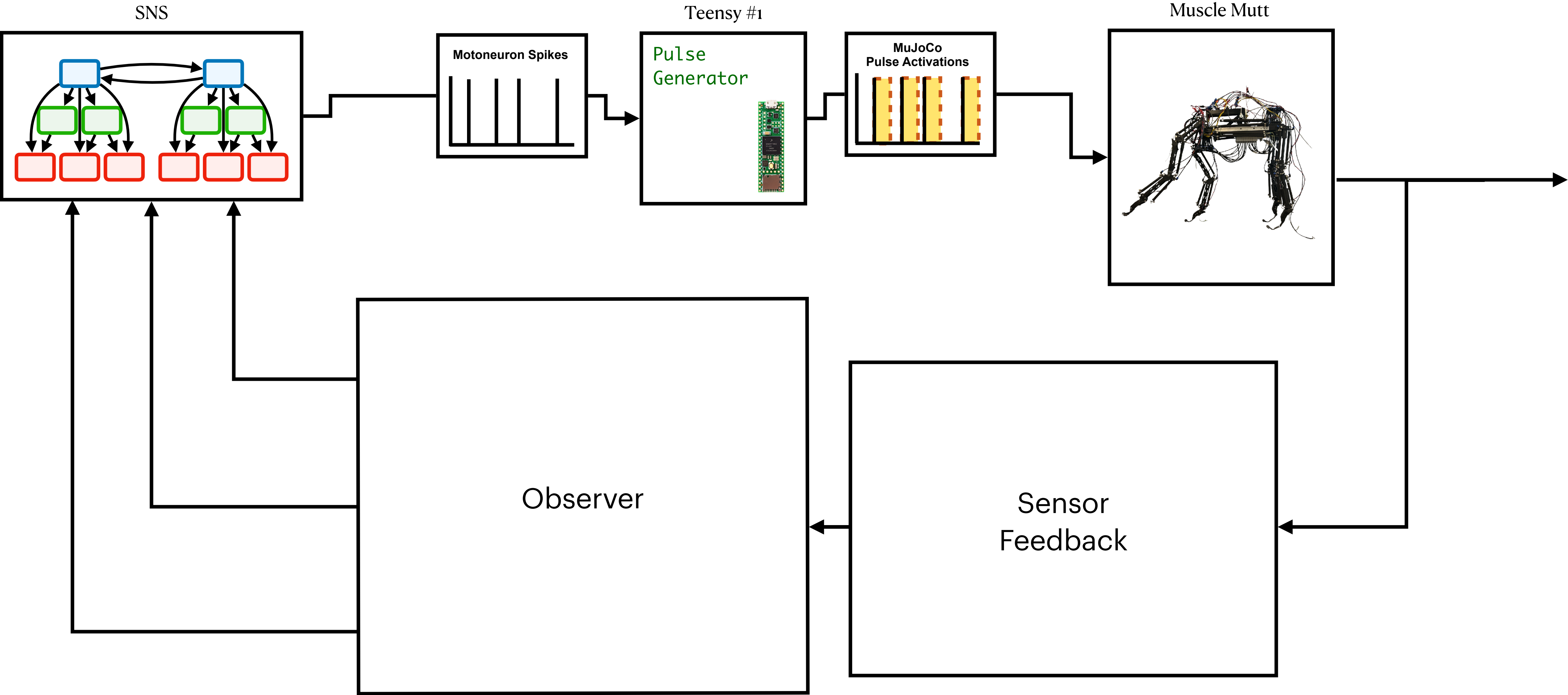
# MuJoCo model simplifies the control loop



# Spiking activations generate air walking in simulation

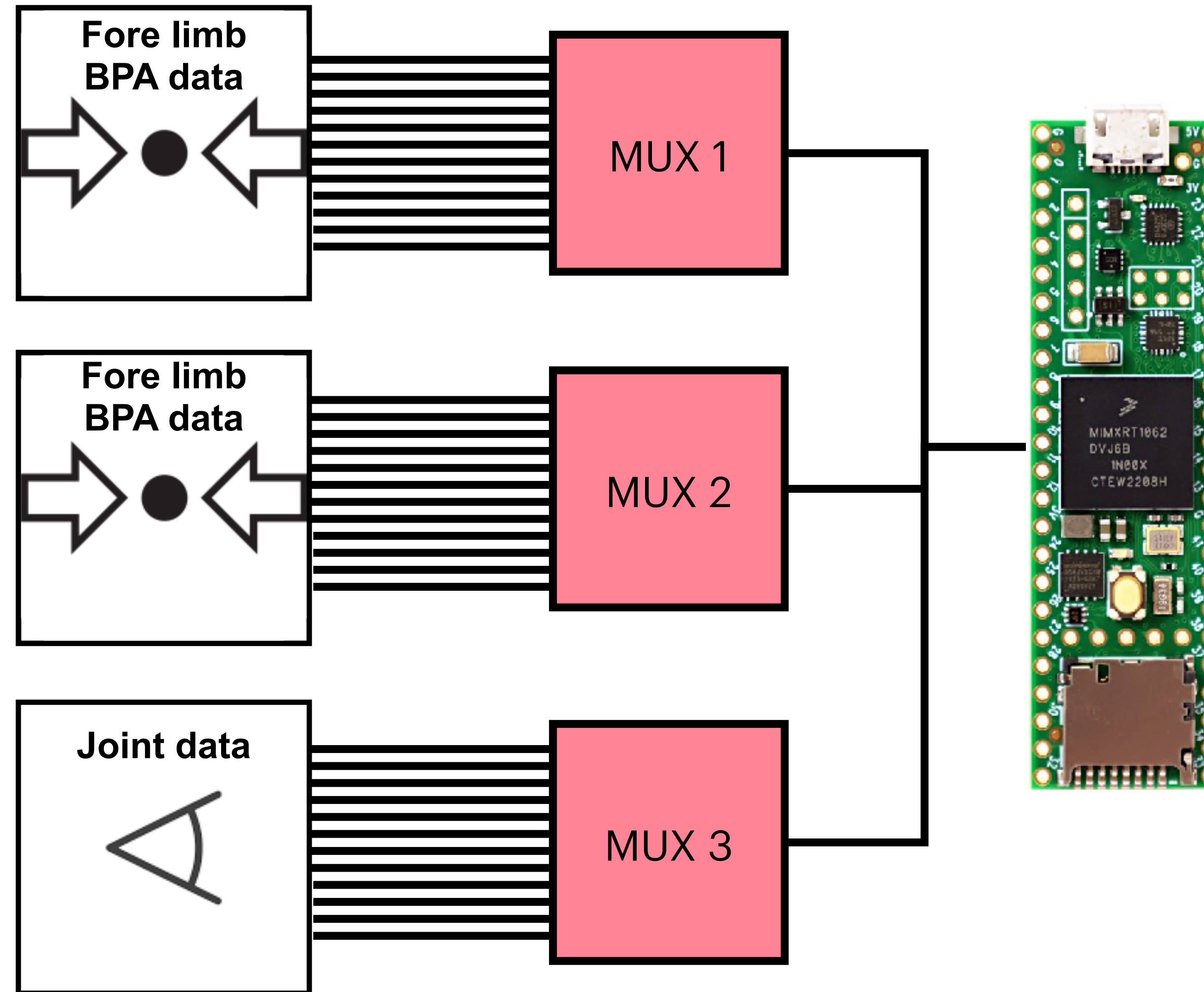


# Sensory feedback is sampled from 24 pressure sensors and 12 potentiometers

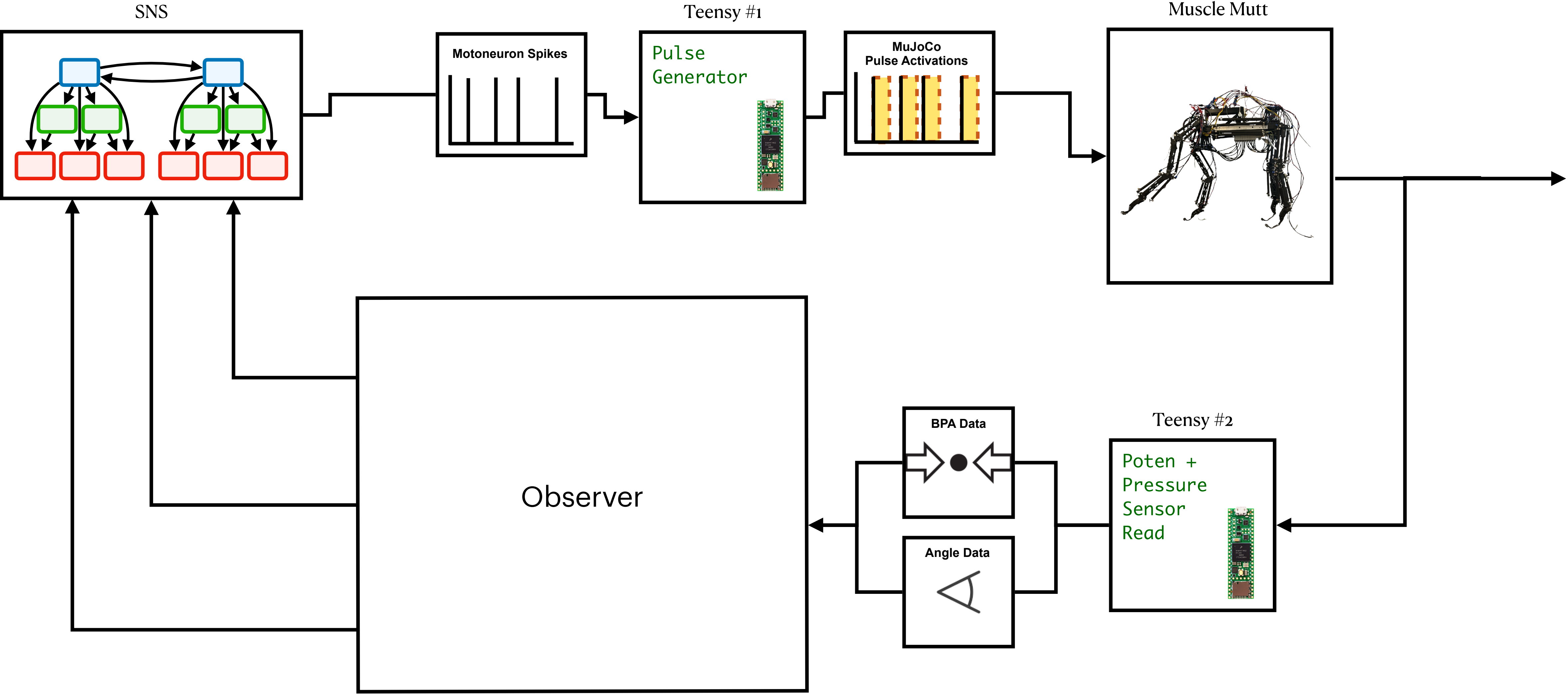




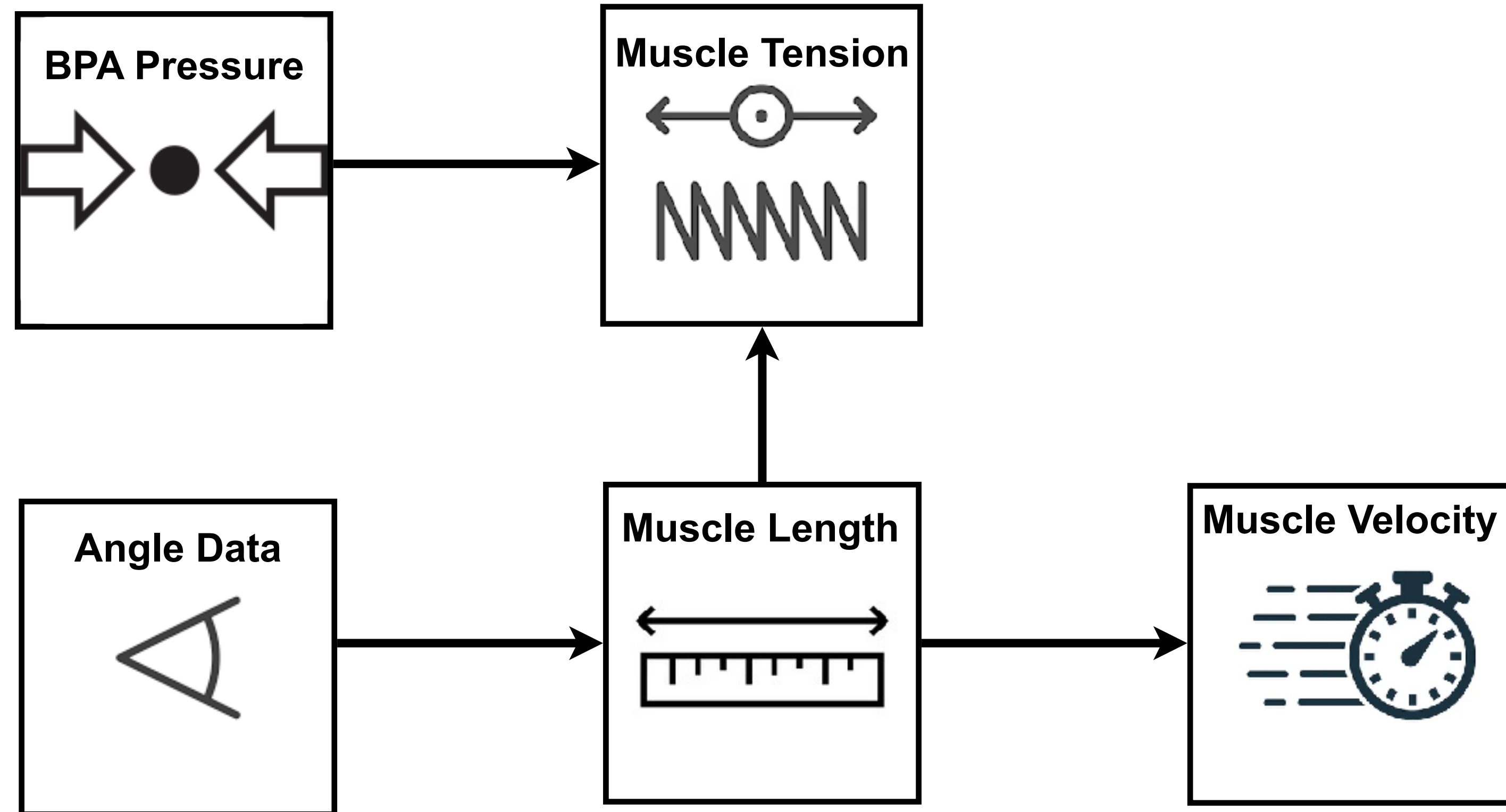
# Data is sent at the spike communication frequency



# Raw sensory data is converted to muscle parameters

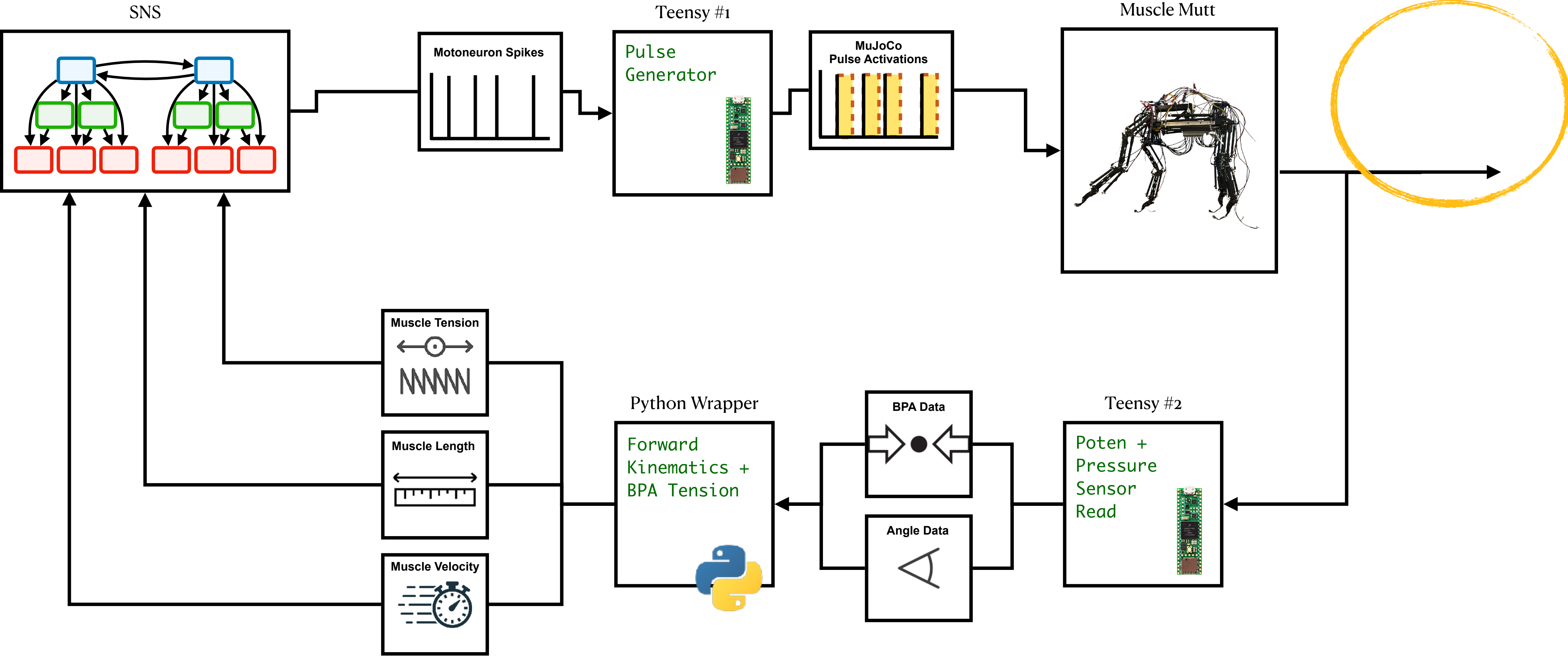


# Muscle parameters are derived using angle and pressure data.



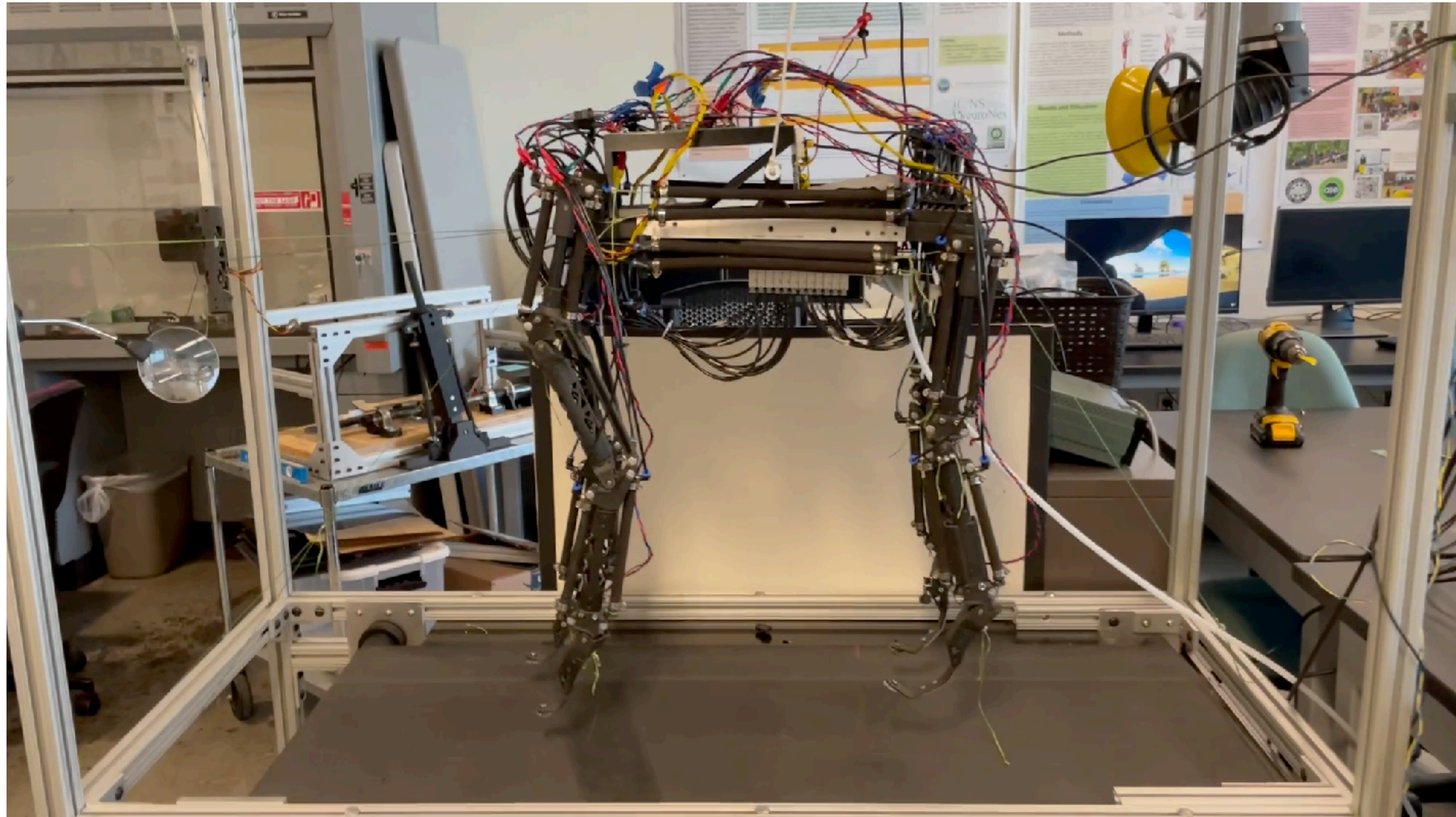


# It's all done! Does Muscle Mutt walk?



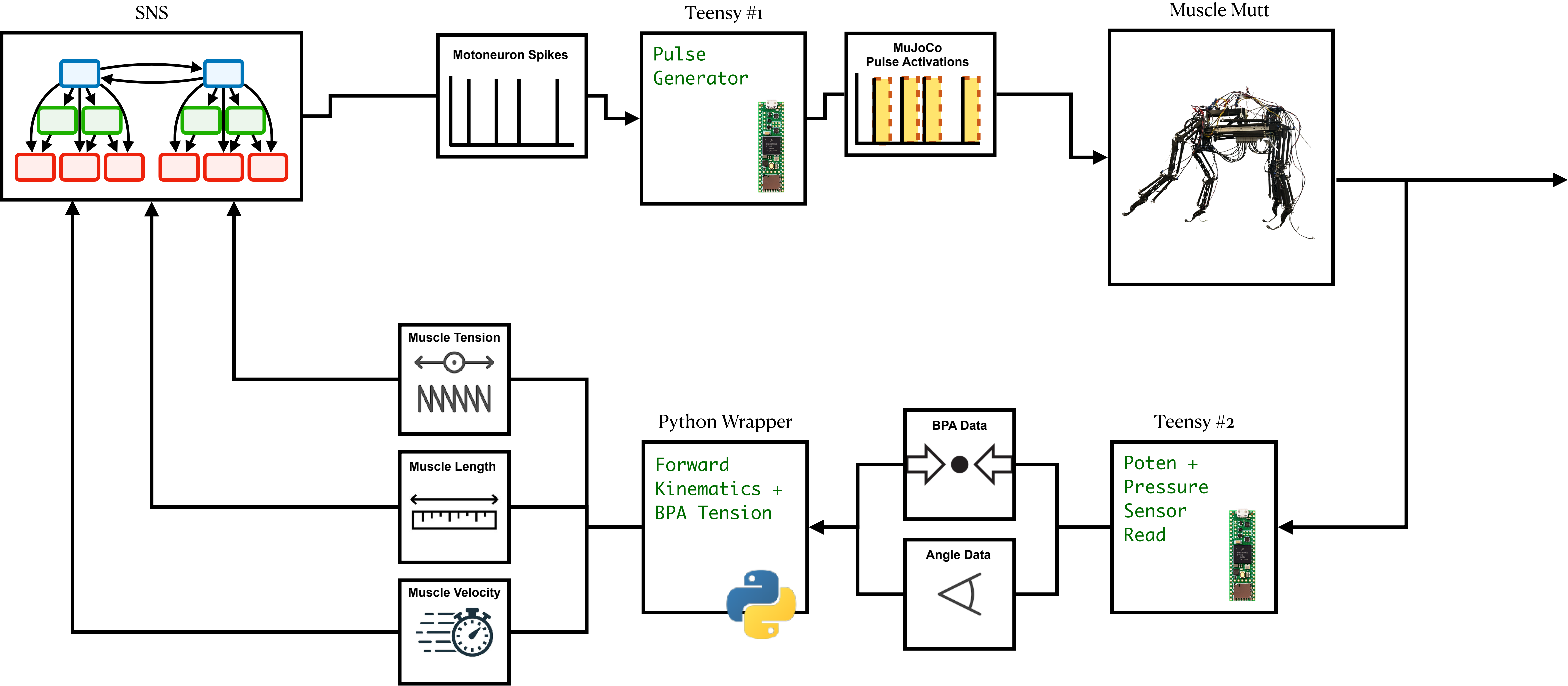


# SNS generates air walking in Muscle Mutt!





# The control scheme effectively connects SNS and Muscle Mutt



Thank you.

